

WHY WE LOVE MUSIC



CARL E. SEASHORE

WHY WE LOVE MUSIC



CARL E. SEASHORE

www.pgdp.net

The Project Gutenberg eBook of Why We Love Music

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

Title: Why We Love Music

Author: Carl E. Seashore

Release date: November 30, 2021 [eBook #66855]

Most recently updated: October 18, 2024

Language: English

Original publication: United States: Oliver Ditson Co., 1941

Other information and formats: www.gutenberg.org/ebooks/66855

Credits: Guus Snijders, Tim Lindell and the Online Distributed Proofreading Team at <https://www.pgdp.net> (This book was produced from images made available by the HathiTrust Digital Library.)

*** START OF THE PROJECT GUTENBERG EBOOK WHY WE LOVE
MUSIC ***

The few minor errors, attributable to the printer, have been corrected. Please see the transcriber's [note](#) at the end of this text for details regarding the handling of any textual issues encountered during its preparation.

Corrections are indicated using an underline highlight. Placing the cursor over the correction will produce the original text in a small popup.

Why We Love Music

BY
CARL E. SEASHORE

Professor of Psychology
and
Dean Emeritus of the Graduate College
The State University of Iowa



OLIVER DITSON COMPANY
THEODORE PRESSER CO., DISTRIBUTORS
BRYN MAWR, PENNSYLVANIA

Copyright MCMXLI by Oliver Ditson Company
International Copyright Secured

Printed in the United States
of America

PREFACE

The magazine, *Time*, commenting on my new book, *The Psychology of Music*^[A], spoke approvingly of the scientific contributions to music, but gibed that, "psychologists have not explained why we love music." As a reply to that I wrote a note, *Why Do We Love Music?*^[B] This seemed to call for a wider excursion implementing the views there taken, as in the present Chapter I. That, in turn, led to the writing of the remainder of this volume, in which each chapter deals with some of the salient factors involved in the development of feeling for music.

These ventures digress from my habitual style of writing, as a technical psychologist, in that I frequently indulge in generalizations and predictions in a practical and popular vein. The attempt to interpret and evaluate present tendencies in the region of forward movements in music naturally takes me into unexplored territory and will stimulate questioning on the part of promoters of various interest in the field. I hold no brief for infallibility of the positions taken, except to say that they are my convictions at the present moment. While they are not direct reports of scientific experiments in the laboratory or the studio, they may be regarded as an extension or interpretation based upon scientific experiments and observations. It is hoped that they may serve as hypotheses or at least a challenge for investigation and practical trial. The reader must judge what is new and what is true from his point of view. Many of the things I advocate are in an experimental stage and in advance of current prevailing practice. This is the reason for presenting them. My aim has been to tease out those elements in the musical situation which help to make music function in our lives and be appreciated.

The following chapters have appeared in magazines: Ch. I, *The Etude* (brief summary); Ch. II, *The Parents' Magazine*; Ch. III, *National Parent Teacher*, under the title, "On Their Musical Way"; Ch. IV, *The School Review*; Ch.

VI, *The Scientific Monthly*. For permission to bring these chapters into this volume I herewith express my appreciation to the publishers.

CARL E. SEASHORE,
Iowa City, Iowa,
March 1, 1940.

[A.](#) McGraw-Hill, 1938

[B.](#) Music Educators Journal, Sept. 1938

CONTENTS

<i>Chapter</i>		<i>Page</i>
I	Why Do We Love Music?	1
	The Musical Medium	2
	Organic response, 2 ; Sounds in themselves, 3 ; Music proper, 3 ; Music with words and action, 3 ; Symbolism, 4 .	
	The Musical Motives	4
	Musical knowledge, 5 ; Musical feeling, 5 ; Musical action, 6 ; Music as play, 7 ; Musical imagination, 8 ; Who loves music?, 8 .	
	Thought Review	10
II	Music Before the Age of Six	12
	From smile to music, 12 ; Music in play, 14 ; Environment, 15 ; Music and speech, 16 ; Musical talent, 17 ; Musical education, 18 .	
	Thought Review	19
III	Music Between the Ages of Six and Ten	21
	A broadened conception of music, 22 ; The analysis of talent, 23 ; Group instruction, 24 ; Formal lessons delayed, 25 ; A sympathetic listener, 26 ; Music lovers vs. virtuosi, 27 .	

	Thought Review	<u>27</u>
IV	Music and Youth	<u>29</u>
	Youth, The Age of Music	<u>29</u>
	The emotional age, <u>29</u> ; The age of serious play, <u>30</u> ; The age of decision and eliminations, <u>30</u> ; The educational age, <u>30</u> ; The age of leisure, <u>32</u> .	
	Music For Youth	<u>34</u>
	Music, an academic subject, <u>34</u> ; Orientation in the grades, <u>35</u> ; Group activities in voice and instrument, <u>35</u> ; Contests, <u>36</u> ; The hearing of music, <u>37</u> .	
	Thought Review	<u>39</u>
V	The Musical Temperament	<u>41</u>
	Physiological irritability, <u>43</u> ; Tonal sensitivity, <u>43</u> ; Artistic license, <u>44</u> ; Ear-mindedness, <u>45</u> ; Affective response, <u>45</u> ; The esthetic mood, <u>46</u> ; Exhibitionism, <u>46</u> ; Symbolism, <u>47</u> ; Precocity, <u>47</u> .	
	Thought Review	<u>48</u>
VI	Musical Inheritance	<u>50</u>
	Essential Premises	<u>50</u>
	Psychophysical Measurements	<u>54</u>
	Thought Review	<u>59</u>
VII	The Future of Musical Instruments	<u>62</u>
	Possible Lines of Development	<u>63</u>

	The improvement of existing instruments, 64 ;	
	New substitutes for existing instruments, 65 ; New ensembles, 66 .	
	New Music	67
	Playing	69
	Specifications for Instrument Construction	70
	Pitch, 70 ; Loudness, 70 ; Time, 71 ; Timbre, 71 .	
	Thought Review	72
VIII	Praise and Blame in Music	74
	Vantage Grounds	75
	Artistic insight, 75 ; The scientific attitude, 76 ;	
	Terminology, 76 ; Musical talent, 77 .	
	Parties Concerned	77
	The pupil, 77 ; The teacher, 78 ; The critic, 79 ; The public, 80 .	
	Thought Review	81

Chapter I

WHY DO WE LOVE MUSIC?

Why does a person love his sweetheart, his food, his safety, his social fellowship, his communion with nature, his God, approaches to the ultimate goals of truth, goodness, and beauty? The answer to each of these is a long story, involving not only common sense and scientific observation but a profound intuitive insight, a self-revelation. In all, it will be found that love is a favorable response, a reaching out for the satisfaction of a fundamental human need, an effort to secure possession, and a willingness to give an equivalent, indeed a more or less unconditional surrender.

In all efforts to describe and explain, we reach out for specific reasons or at least rationalizations. Modern science has made great strides in revealing and describing all sorts of reasons for such emotional experience and behavior. The theory of the evolution of man, the anthropological implementation of this in the history of the rise of mankind, the psychology of the mental development of the individual, the comparison of this with animal behavior, and the inspired interpretation of these motives in literature, especially biography, autobiography, and poetry, are sources to be drawn upon. We have the adage that the explanation of one blade of grass involves the explanation of all the forces of nature. This aphorism certainly applies in the attempt to explain any particular human love.

It is therefore evident that any attempt to account for a specific affection, such as the love of music, must be fractionated, placing responsibility in turn upon the scientist, the artist, and the self-revelation of the inspired music lover at each culture level. It has become the recognized function of the psychology of music to integrate the contributions from all scientific sources, such as anatomy, physiology, anthropology, acoustics, mental hygiene, and logic, in their bearings upon the hearing of music, the appreciation of music, musical skills, theories, and influences. To account for the emotional power of music, the psychologist must consider the

taproots of the artistic nature of the individual in relation to the nature of the art object, music. He must trace the unfolding of the organism as a whole from inherited reflexes, instincts, urges, drives, and capacities in an integrated pattern; he must consider the function of the art in human economy and especially the goals attained by the pursuit of the art. In this task there is room for intricate specializations and division of labor. It is my purpose here to present merely a rough skeletal outline of some of the outstanding features which underlie the love of music from the psychological point of view.

Every impulse has two aspects: attraction and repulsion. All of us love music in some degree; all of us hate some music; and most of us in the economy of nature are comparatively indifferent and extravagantly wasteful to the role that music might play in our lives. Hatred and indifference to music are important realities in life worthy of serious consideration; but our topic restricts us to the positive side of musical response, the love of music.

THE MUSICAL MEDIUM

Organic response. Man is born with a psychophysical organism which registers sounds and responds to them somewhat like a resonator, which selects, amplifies and aids in the integration of auditory impressions. Our whole organism responds to sound involving the central and peripheral nervous system, all the muscles, all the internal organs, and especially the automatic nervous system with its endocrines, which furnish the triggers in the physical generation of emotion. Experiments from various sources have shown that sound acts physiologically on nervous control, circulation, digestion, metabolism, body temperature, posture and balance, hunger and thirst, and in general, the groundwork of pleasure and pain. The physical organism as a whole responds to sounds in specialized functions.

Thus, man comes into the world tuned to music. The organism responds to sounds from earliest infancy. Back of all conscious awareness, back of all musical feeling, even back of subconscious assimilations and elaborations is the purely physiological response which is a function and a condition of well-being. This physiologically beneficent response of the organism to sound underlies all musical experience; without it we could not love music.

Sounds in themselves. Like colors, sounds may be beautiful in themselves, quite apart from music. A single sound in nature or art is capable of appearing in endless variety in terms of pitch, dynamic value, duration, tone quality, and noise. It may be an object of beauty in itself in thousands of ways quite apart from its utility in music or musical perception. We find the tonal world in which we live full of beautiful and useful sounds which we love because we are capable of intellectual and emotional response to their beneficent influence. They play a large role in our feelings of attraction and adjustment. They may be beautiful to the untutored and intuitive mind as well as to the intellectually and esthetically cultured mind in the same way that flowers may seem beautiful to a child because they arouse an immediate pleasurable feeling; and yet they are not music but merely the raw material from which music is made. These raw materials from which the musical structure is raised are themselves beautiful, quite apart from musical experience or behavior. They play a large role in the love of nature.

Thus, before the beginnings of music, primitive man responded affectionately to the sounds of nature and was guided by them in his daily life. Even before language took form, single sounds carried meaning and gave satisfaction. Man took pleasure in his own vocal utterances or mechanically produced sounds which played a large role in his human economy and development.

Music proper. Sounds may be woven into beautiful patterns. This is music. We admire the melodic progressions, the rhythmic patterns, the harmonic structure, and the qualitative modulations in the flow of beautiful sounds. Harmony, balance, symmetry, contrast and fusions become embodied in musical form. Here the object of our affections is the artistic creation. The place of the musician is quite analogous to the astronomer's feeling of the sublime as he looks into the heavens in the light of his knowledge of the nature and movement of heavenly bodies.

Music with words and action. Much of the charm of music lies in its association with words which carry the message, as in song. The center of interest in much of the vocal art lies in the meaning conveyed by the words where the music serves as an artistic embellishment. This is true of the lullaby, the cowboy song, the lover's plea, and grand opera. Likewise, much of the charm of music lies in its association with overt action as in dances, work songs, marches, and games, where action is rhythmic. This added power of the music lies not only in the dance steps but more conspicuously in the suggestive rhythms divided into intricate patterns often far surpassing the score or the physical performance. That is what we mean when we say music carries. In such situations the musical appeal may lie for one person in the verbal message or the overt action and for another, purely in the musical appeal. Yet both words and action on the one hand and music on the other are enriched through the association.

Symbolism. Music finds its highest and most universal expression in symbolism. Music is primarily a way of expressing moods, attitudes, feelings, and longings in generalized form. The listener tends to live himself concretely into the feeling suggested. In the esthetic mood he is not aware of the mechanics of the symbolic suggestion, for which the art has many resources, and he may not be aware of the music as such; but he lives realistically within his own personal realm of interests. Thus, music sounds the keynote on great festive occasions in the powerful forms of festive music, as in the great sacred oratorios and simpler but beautiful forms of

church music or in triumphant marches and other festive celebrations in major form. But minor forms, as in tone poems and haunting melodies, work on the same principle and perhaps fully as effectively. From the grandeur suggested by the sonata to the serenity arising from the simplest bit of improvisation in voice or instrument, music has unlimited power to seize the individual for some form of dreamlike realization of the subjects of his longings.

THE MUSICAL MOTIVES

What we are called upon to explain then in the attraction for musical art is essentially the motives which drive man to the creation, appreciation, and performance of music. One of these motives is the love of knowledge as a thing in itself, the understanding of what is, and the power of passing from vantage ground to vantage ground in the logical creation, appreciation, and execution of art forms.

Musical knowledge. This love of music for its cognitive value can be traced from the earliest musical achievements, as in the growing acquaintance with song, sight reading, qualifications for participation in music, and appreciation of art forms, throughout all stages in the musical development of the individual up to that of the highest interpreters and creators of music. While music is a play on our feelings and appeals primarily to our emotional life, an intellectual mastery of the process, the ability to understand artistic meanings, the ability to construct beautiful art forms, the ability to analyze elements in the power of music, the ability to see the relation between musical art and other forms of art, and the ability to comprehend the unity of all the arts, are basic in our love of music. Even in the cool and logical pursuit of the science of music, foundations are laid for the deepening of insight and the revelation of artistic values. Glimpses into the vistas of unexplored resources intensify the admiration, the feeling of awe, the glimpse into the infinite which is love of the object pursued.

The role of intelligence in music is well illustrated in recent experiments in which vocabulary was measured in three groups; namely, ten nationally well-known composers, ten of the most successful students in a large class in composition, and ten of the least successful students in the same class. It was found that the master composers and the successful students of composition ranked in or near the top in a test of general vocabulary; whereas, the unsuccessful students ranked near the bottom. Since knowledge of words is an index for the possession of ideas, it is significant to note that successful composers are persons who have a large and discriminating command of ideas.

Musical feeling. It must be recognized that the love of music is essentially an unanalyzed feeling. Countless people feel the esthetic appeal

in music without understanding anything about it. It may be like the notorious puppy love, which is frequently blind, but nevertheless a deep love. This is particularly true in the earlier stages of the development of musical interests. But it is occasionally in evidence in the successful singer on the stage who may be blissfully ignorant of the principles underlying his art, the media he seeks to mold, or the significance of his message. There is much justification for the performer's forgetting what little he knows and indulging in self-expression in a state of abandon in which he deeply feels his message and expects to convey this feeling to the listener.

We must distinguish between two attitudes in listening to music and in the performance of music: the critically analytical and the purely emotional. An intelligent musician is capable of both and loves both. In the learning stages he pursues the former attitude primarily until techniques are mastered and habits are formed which operate automatically in the musical situation. This is also the dominant attitude of the music critic. But in seeking the enjoyment of music and in the unified expression of a thing beautiful, the musician takes the other attitude. Paderewski would be hopelessly lost and ineffective if, at the moment of performance, he should be consciously aware of all the art forms of which he is master. The successful performance comes in an inspirational attitude, the semi-ecstatic feeling of the beauty one seeks to convey, a state of forgetfulness of self and concrete facts.

Thus music is a language of emotion. Through it the composer and the performer convey their own emotions to the listener. It is a message and a means of communication which enable the performer and the listener to live for moments in the same tonal world of pleasure. Our muse is jealous and seeks to exclude all intruders at the moment of her artistic appeal.

Musical action. On a par with the intellectual and emotional approach is the role of action in music. Consider for a moment the central place of rhythm. The composer presents a hierarchy of rhythms: the measure rhythm, the phrase rhythm, the sentence rhythm, the movement rhythm, all moving into a unified beautiful artistic structure. The performer takes this as a cue and adds or detracts, as the case may be, by his personal interpretation. Modern psychology has shown that all musical listening is action, a constructive response on the part of the listener.

All rhythm is primarily a projection of personality. My rhythm flows from what I am. A large part of the pleasure in music comes from a

satisfaction in what rhythm does. Rhythm facilitates perception by grouping; rhythm adjusts the stream of attention; rhythm gives a feeling of balance; rhythm gives a feeling of freedom, luxury and expanse; rhythm gives a feeling of power, it carries; rhythm, as in the dance, stimulates and lulls, contradictory as this may seem; rhythm finds resonance in the whole organism; rhythm arouses sustained and enriched associations; rhythm reaches out in extraordinarily detailed complexity with progressive mastery; the instinctive craving for experience in rhythm results in play, which is free self-expression for the pleasure of expression; rhythm plays not only with temporal but also with dynamic and qualitative aspects of tone. Subjectively, rhythm in music is a play within a play: The composer anticipates it, the performer gives the cue, and the listener expresses himself in it.

Music as play. All art is play, and the charm of music, the purest form of art, lies fundamentally in the fact that it furnishes a medium of self-expression for the mere joy of expression and without ulterior purpose. It becomes a companion in solitude, a medium through which we can live with the rest of the world. Through it we express our love, our fears, our sympathy, our aspirations, our feelings of fellowship, our communion with the Divine in the spirit of freedom of action.

Note the fundamental characteristics of play and observe how in these lies the power of attraction in music as play. Play gives a feeling of self-realization; it is the experience of growth. It expresses the racial life and in many respects is a reversion to type: It has been said that we are all of the same age—millions of years. Play is a realization of the sense of freedom; it attracts, engages and fascinates by the very satisfaction which it engenders and which supports it; the dance, when it is real play and not mere social labor or conformity, carries the dancer in so far as he falls into a state of diffuse and dreamy consciousness, intoxicated by the sense of pleasure, lulled by the automatic rhythmic movements, and soothed by the melodious and measured flow of music. Play gives satisfaction in the feeling of being a cause, of having creative power. Play is essentially social and finds its highest realization in good fellowship. Play is positive, an expression of the joy of life. The unrestraint and spontaneity of play result in strenuous and whole-hearted exertion; the seriousness of play is one of its fascinations. Success in play lies in its fictitious nature; it rests upon make believe;

liberated from realities, it accepts the ideal and lives it as real. In the possibility of playing with the ideal lies a fundamental charm of music.

Musical imagination. Music is by no means limited to what is composed, performed and listened to in the objective situation. Its main field of operation lies beyond the sensory impressions and overt actions. Its principal domain is the tonal world of memory, imagination, thought, and pure feeling. Millions of people are today under the spell of *Over the Rainbow*, as rendered in *The Wizard of Oz*, a simple, compelling thing which takes possession of us in the dream, and in the humdrum of daily activities; it lives within us realistically, quite apart from actual sounds so long as it is novel. This is especially true of the higher forms of art with all their intricacy and refinements in artistic form which the trained musician can re-live or create. *Over the Rainbow*, the expression of our freedom and self-realization in the spirit of adventure, lies within the power of music.

Who loves music? The love of music is not universal. Deep, warm, and poised devotion to music is comparatively rare. Much of music is plain work, sheer drudgery. Much is climbing toward a goal never to be attained. Many who ply the art of music can hardly be said to love it in the long run. There are aspiring artists who devote a lifetime to the mastery of the skills, but become hypercritical and sour when they fail to feel the esthetic glow or gain the command of public acclaim. Many an aspiring amateur suffers a similar defeat. To the masses, music is but fleeting incidents, occasional whiffs of the overflow from the wealth of human appeals to a latent artistic nature. It is a notorious fact that many who profess a love of music do not have it, but are mere pretenders and imitators, conscious or unconscious, and that many who disavow it are merely dying with all their music in them.

For such failures and inadequacies there are many possible explanations. One of the impediments to the love of music is the absence of the "gift", a naturally musical mind. In this there are enormous differences in kind and degree. It follows that there will be corresponding differences in the kind and degree of love for music. While music springs spontaneously in the gifted child and youth, education is as essential for music as it is for science or language. One might as well attempt to acquire learning without study as to acquire music without training. Then again, as is a person's intelligence, so is his music. And creative imagination is a tool with which music is fashioned from childhood to the heights of artistry. Furthermore, we must

recognize that for really expressive love and devotion to music we must look to the often justly or unjustly maligned musical temperament.

Yet the company of music lovers is great. Music is the most universal avocation. This has been true of all races throughout all times and at all culture levels. Only a fraction of one per cent of persons who hear music or practice it do so vocationally. We, the people, preserve it primarily as an avocation, an activity purely for pleasure and cultural enrichment. The love of music abounds at our time and in our country; yet we are but at the beginnings of a dawning musical era. The increase of leisure time, high educational level, and the astounding invention of instruments for the production and transmission of musical sounds forecast its rise.

Why then do we love music? Among other things we love it because it creates a physiological well-being in our organism; it is built from materials which are beautiful objects in themselves; it carries us through the realms of creative imagination, thought, actions, and feelings in limitless art forms; it is self-propelling through natural impulses, such as rhythm; it is the language of emotion, a generator of social fellowship; it takes us out of the humdrum of life and makes us live in play with the ideal; it satisfies our cravings for intellectual conquest, for isolation in the artistic attitude of emotion, and for self-expression for the joy of expression.

THOUGHT REVIEW

General Principles

- (1) Music is pursued primarily for the pleasure of the pursuit without ulterior purpose.
- (2) When music renders service, this service is essentially the giving of pleasure.
- (3) The depth and the quality of a person's affection for music vary significantly with the degree and kind of musical inheritance.
- (4) If the natural urge toward music is there, it will be readily molded through training and facilitation of the environment.
- (5) It is psychologically possible to present a natural history of the origin and development of the love of music in terms of its objects and motives.
- (6) The inceptive psychology of musical experience and behavior can be pursued in the laboratory and the studio through experiment and measurement.
- (7) There is a charm in knowing *that* we love music but even more so, *why* and *how*.
- (8) It is also worth knowing why some people *do* not and others *cannot* love music.

Questions to Consider

- (1) Is the play attitude in music opposed to hard work?
- (2) Why is high intelligence essential to the composer?
- (3) Can a moron love music genuinely?
- (4) Is the musical temperament essential for the love of music?

- (5) How can the adrenal glands affect the love of music?
- (6) Is the love of rhythm primarily inherited or acquired?
- (7) What hereditary factors may block the development of the love of music?
- (8) What factors in musical talent are most telling for the development of the love of music?
- (9) What educational motives are most effective as a means of enhancement of the love of music?
- (10) What is the nature of musical ecstasy?

Discuss These True Situations

- (1) The typical "house" for the season's performance of the symphony orchestra is arrayed in festive attire and poses in attentive and festive mood. Make an estimate of the relative proportion of the audience there (a) to see and be seen, (b) to perform a social obligation, (c) to satisfy curiosity, (d) to learn something about music, (e) to thoroughly enjoy the orchestral music.
- (2) A primitive savage lives close to nature and observes sounds produced by man and beast, by wind and sea. May he feel musical affection for these sounds apart from their utility and apart from knowledge of man-made music?
- (3) In an aristocratic family there are four children, all manifestly lacking in musical talent, but intelligent and co-operative in education. They are all given excellent advantages of musical education and environment and pursue them as a matter of course. Are they likely to come into a warm love of music?
- (4) It has been observed that when one listens to very beautiful music he tends to swallow saliva by gulps, as the young lover does in a state of infatuation; and that when he hears vile music, there is a tendency to want to spit. In other words, there are two ways of disposing of saliva

under emotion: Attraction, indicated by swallowing; and repulsion, indicated by feeling the need of spitting. Can you verify that?

Chapter II

MUSIC BEFORE THE AGE OF SIX

The psychology of music and the psychology of the child are giving us new vital conceptions of the nature and role of music in child life. To understand this fully is to understand adequately the nature of the child mind and the nature of music.

From smile to music. All mental development begins with some inherited form of behavior and gradually differentiates into richer and richer meanings and forms of expression. The taproot of all music is the smile. This in its first appearance is a pure reflex, expressing the well-being of the organism.

Observe some typical steps in its history in high lights. When the infant has had its fill from its mother's breast, its head falls back and the mouth puckers as a result of withdrawal from the nipple. The mother looks at this and says to herself, "He is satisfied." The child has thus acquired one means of communication—the expression of well-being. This rapidly radiates into many situations. When the infant is patted on the back, is bathed, is rocked in the arms, or feels the waft of comfortable air, the same puckering of the mouth seems to convey to the mother a sign of well-being, and the meaning of this puckering is thus enriched.

It gradually radiates from the lips through more general expressions of comfort in the face as a whole, and we have a clearly developed smile. Gradually it becomes associated with sounds—inceptive gurgling, simple droning, light chatter, and other inceptive forms of vocalization, always accompanied with a smile, which thus develops new meanings, and mother and child acquire mutual understandings, because from the first the mother tends to respond sympathetically in like language.

As this association grows it takes the form of audible laughter which, at a comparatively high stage of development, becomes a "ha-ha-ha" with musical inflections. This is a form of language, still not expressing specific

ideas, but the general attitudes of well-being, comfort, satisfaction of young life as a whole. The mother knows it, loves it, and responds accordingly; and this mother's response draws from the infant a growing hierarchy of new types of responses, meaning the same thing.

Gradually these infantile sounds develop inflections and modulations in pitch, in loudness, in rhythm, and in tone quality. These inflections are the beginning of beauty in voice, and each new conquest gives a new form of satisfaction to the child and mother.

The mother may stimulate natural forms for expression of mutual feeling by her own musical laughter. Gradually the playing with these sounds becomes an object in itself, the making of a particular sound for mere pleasure. This is the beginning of singing and the appreciation of musical sounds.

We observe the child playing with modulations in pitch, in loudness, in duration, and in different kinds of tone quality. This is the beginning of musical experiment, of musical creation, and musical appreciation in the child. Blessed is the mother who can appreciate the music thus born. She then sees the significance of the rhythmic patty-cake as music which gives pleasure to mother and child alike. She then begins to understand that the jingling of a bell is sweet music to the child. She can see how the appreciation of rhythm is gradually revealed through the progressive development of means of making sounds and hearing varieties of sound. Perhaps unconsciously she sets patterns for a musical inflection which the child begins to imitate. To the child, noise is music, and the discovery of noises and the mastery of various noises play an immensely rich and important role, even in the highest forms of adult music.

Thus, the recognition and feeling of pleasure in sounds, and the power to make agreeable sounds, reveal to the child an unfolding musical world. His whole organism responds to it. This is an element of musical feeling.

Let me digress for a moment to say that the beginnings of mental life tend to develop from two fundamental needs: attraction and repulsion, likes and dislikes, approaching and getting away from—these being respectively the positive and the negative aspects of adjustment. The smile is the sign of the positive side. The companion piece of the smile as a taproot is the frown, which gradually develops through crying and is the reciprocal of laughter. The history of the frown and crying is quite parallel to the history

of the smile and laughter in mental evolution and in the development of the individual.

Music in play. While it does serve a purpose as language in social adjustment, the musical activity of the young child is expression for the love of expression itself. This play aspect early differentiates itself from the use of sounds to convey meaning in language.

The child composes, rapidly revealing new melodic progressions, new rhythms, new kinds of sounds, and new patterns in the durations of sounds for his own delight in self-expression. Through them he wins manifestations of appreciation from those around him who constitute his audience. He repeats each new achievement as the momentary goal of play until new patterns progressively take their place. In these actions of musical composition, the same mental faculties that we see in the active adult composers are at work, but limited to the child's natural level of successful achievement.

He early reveals command of the four elements of all music, as such; namely, pitch, loudness, time, and kind of tone or tone quality; and we observe the unfolding of melody, dynamic expression, rhythm, and richness of tone. Through these he imitates the sounds of nature, speech, music, and noises at his level in the surroundings. In terms of these he develops memories, indulges in fantasy and creative imagination, and gradually begins to think about music. These are the avenues through which the child expresses his needs and urges vocally or instrumentally; but they are music in so far as they are indulged in for the pleasure of the hearing or the making of the sounds themselves.

Note that the child composes by performing; therefore, each musical form that he develops is clearly at his command in performance. Notice also that the repetition of achievement is limited by the play attitude of always demanding something more difficult. Composition, performance, appreciation, body response form interlocking steps. That is what makes the procedure natural. The joy of conquest, characteristic of play, is the dominating motive. Here nature has her way in the development of knowledge, appreciation, and skills.

Musical activity is normally a form of play, expression for the satisfaction in the expression itself without ulterior motive, and this attitude may be carried through life. A person who cannot take a play attitude toward music perhaps has no music in him. The play attitude does not free

one from effort, even systematic and arduous effort, in the acquisition of the art. Witness all the sports at all stages of growth. There is nothing to relieve us from understanding facts involved in music; but the driving motive is found in the play attitude, and the result is pleasure in play. On these observations of nature's ways in play, the future pedagogy of music will be built.

Environment. It is astonishing that the child is often treated as unmusical unless he can sing or play adult compositions or show an intelligent appreciation of high art forms. How pitiful it is for a mother to say that her child is not musical because he does not sing her songs and understand her artistic playing. How vastly could a mother's appreciation of the child be increased if she realized what constitutes music at his level and how fundamental the musical reactions at his level are to the development of music in the adult!

To be musical, the child must be musical in response to his environment. There are natural laws of evolution in the race and in the development of the individual for types of reactions to the music that abounds around us in nature and for the various means at our command of expression through imitation of them. To the primitive tribe, the drum is a powerful, thrilling musical instrument. So are all forms of drumming to the child. He imitates the whistling, tooting, rattling, banging sounds in his environment, sometimes until he becomes noisily tiresome. He feels in harmony with the clock that ticks, the birds that sing, the dog that barks, the cat that mews. He loves to bang on the piano and blow his horn.

This craving for pleasure in sounds radiates through the sense of rhythm into graceful movements, the beginnings of dancing and dramatic action, even from the crudest rhythmic kicking and tapping movements of the infant. His speech becomes rhythmic, melodic, dynamic, beautiful. His whole body becomes reverberant in response to the sounds of nature. Laughter progressively acquires new and beautiful forms. Even crying may give satisfaction of an artistic sort. The swinging of the pendulum of the old clock on the wall is music. The patter of rain, the splashing of water have musical elements.

The child does not think of the artistic forms as does the musician; but like the canary which, even if grown in isolation in a soundproof cage, in due time produces his natural songs in repertoire, the child instinctively

comes out in melody, dynamic modulation, and rhythm. But all these are modified by the environment.

Music and speech. Speech has the same media as music; namely, pitch, loudness, time, and timbre which result in such musical forms as tonal and dynamic inflection, rhythm, articulation, and vowel quality. A child is, of course, not conscious of any of these as such, and yet, under favorable circumstances, will quickly develop beautiful speech, which means that it is well inflected, well modulated in loudness, beautifully rhythmic, rich and clear in vowel qualities. If the child has a good ear, instinctive liking for these aspects of speech will develop surprisingly early.

To give the child musical environment means therefore not only exposure to formal music but rather a motivation for hearing musically all sounds around him, for acting rhythmically, and feeling the rhythmic impulse in all forms of activity, for responding by imitation or other forms of appreciation to all sounds beautiful. These acquisitions naturally take the form of beautiful speech. Musical education in the nursery, therefore, comes most effectively through informal education toward beautiful speech.

The child becomes proficient in speech long before he becomes correspondingly proficient in musical performance and appreciation. It is therefore very important to recognize that music and speech employ the same medium; namely, sounds which vary in pitch, loudness, duration, and kind. The child reveals flexibility, richness, rhythm, and all other forms of meaningful inflection in speech earlier than he does in music. Indeed, by the time the child leaves the nursery, even at the age of five, his characteristic form and command of speech are fairly crystallized. The command of elements of beautiful speech is the first step in a beautiful singing voice.

Speech is an index to character, and the means for the development of character. Beautiful speech is musical speech. Genuinely beautiful speech is a revelation of beautiful character. Let the mother who worries about early piano or violin lessons first give thought to formal sympathetic cultivation of a beautiful speaking voice.

Unfortunately, the child's speech is very largely determined by imitation of those around him. How few mothers and fathers, how few teachers, how few older children have beautiful speech! The young child as a rule, therefore, encounters unfavorable speech environment. The civilized world

is just awakening to the possibilities and significance of beautiful and effective speech.

Train the young child to the appreciation and development of power in beautiful and effective speech, and you will have laid the best foundations for musical appreciation and formal entrance upon musical training.

Musical talent. Children differ in musical talent, both in degree and in kind. One normal child may be twenty-five, fifty, or one hundred times as sensitive as another normal child in matters of time, in matters of pitch, in matters of loudness, and in matters of tone quality. The normal child may be very high in one of these four musical traits and low in another. Let me illustrate how these traits may be observed very early in a very highly musical child.

Playing with a little girl eleven months old, I noticed that she responded to the music over the radio. I put a simple two-step on the victrola, and she marked the time correctly by a free sympathetic swinging of the arms. I changed this to waltz time, and she picked up the rhythm. As she could not yet stand on her feet, I held her on all fours, and then she shimmied with her trunk. Was that child musical? I could give one positive answer. She had a splendid sense of rhythm and urge for rhythmic action. As I watched her in succeeding years, she very early developed original dances, and at the age of four gave delight in original "shawl dances." Her speech very early was beautifully inflected. Her speech was also very early characterized by fine and meaningful modulations in loudness for emphasis and meaning. She gave early evidence of power to imitate different sounds.

Of course, the less musical a child is by nature, the more difficult it is to find early evidences of this sort. We do not need measuring instruments so much as we need training of teachers and parents to an understanding of what constitutes musical capacities so that we can observe the child critically in his early natural responses. By the age of eight or ten these specific capacities will become more conspicuous, and at that age the competent psychologist in music can analyze and measure talents reliably.

Musical education. When and how should musical education begin? It should begin in the earliest infancy by giving the child a musical environment suitable to elicit his response. This means not simply the hearing of formal music, but, far more significantly, a sympathetic response to the child's natural vocal expressions at each level, even to the making of sounds of all kinds.

The child from the first needs a sympathetic audience. It is not so much how beautifully the mother sings as how sympathetically she responds to the beginning croonings of the infant; and this sympathetic enjoyment includes recognition and encouragement for the hearing of all sounds around, whether animate or inanimate. The mother's first task is to be a good listener.

The first elements of formal musical training should be devoted to speech. Ideally this would come most effectively through the child's opportunity for hearing and imitating the beautiful sounds and speech of those around him. Even if the mother and other associates cannot set the model for the child, they can do a great deal to further musical development by showing their appreciation of the instinctive outcroppings of the musical qualities of the child's speech.

During the first six years there should be no formal musical instruction; but, by the end of that period, the musical child should have gradually acquired a sense of appreciation for musical sounds, pleasure in self-expression in musical intonations, confidence in his ability to compose a tune, some proficiency in singing and good speech, and some degree of satisfaction in free playing with an instrument.

The principles here developed for early childhood have profound implications for later musical education. Let the emphasis lie upon the broadness of the meaning of music to the child, upon the child's learning by doing, at his natural level of successful achievement, and upon the utilization of natural motivation in place of formal instruction.

THOUGHT REVIEW^[C]

General Principles

- (1) Musical activity is always a form of play.
- (2) To be musical the child must be musical in response to his environment.
- (3) To the child noise is musical.
- (4) Musical education for the preschool child comes most effectively through formal education toward beautiful speech.
- (5) Musical education should begin in infancy by giving the child a musical environment suitable to elicit his response.

Do the above statements by the author modify your own theories in regard to the musical education of your children?

Questions to Consider

- (1) What may we be doing when we say to our husky two-year-old, "Oh, Johnny, don't make so much noise?"
- (2) Why is it that children so quickly lose their lovely childish intonations and inflections?
- (3) How does Doctor Seashore endeavor to help us recognize music in our children?
- (4) In our efforts to develop musical ability and response in the child, what errors do we, as adults, commit?
- (5) Is it ever too late for adults to cultivate a beautiful speaking voice?
- (6) Do you agree with Doctor Seashore that "genuinely beautiful speech is a revelation of beautiful character"?

- (7) What methods are you using with your young child to elicit musical responses? Are they in accord with Doctor Seashore's advice?
- (8) Do you think that the radio is helping to encourage musical speech?
- (9) What practical steps can the adults in the family take toward improving their own influence upon the child's musical development?

Discuss These True Situations

- (1) I listened to the droning voice of a mother reading to her small son. Would it be better for the mother to tell the story with animation?
- (2) Mrs. Brown is what is termed "high-strung." She has much to do with household duties. Her three toddlers ask many questions, over and over again. She finds herself speaking louder and louder with each repetition. What would you suggest? (In a similar situation a five-year-old said, "Mother, why don't you relax a little?" C. E. S.)
- (3) In the Black household the radio is on continuously. Is this helping the musical education of the older and younger children?
- (4) "I simply can't get Johnny to practice his music lessons unless I sit down with him and make him do it," says Mrs. Young, who voices the complaint of many ambitious parents. What would Doctor Seashore say?
- (5) A mother, musically talented, said to me, "I simply can't help my children at all with their music. I get frantically impatient with them."
- (6) In a crowded railroad station I was admiring a lovely looking woman who held an equally lovely looking child by the hand. Then she spoke to the woman sitting next to me and it was with great effort I kept a look of consternation from appearing on my face. Her voice was harsh, metallic, unmodulated. Is it possible for her to help herself?

C. This outline was prepared by May E. Peabody, Supervisor, when it appeared in *Parents' Magazine*. It seemed to me so

stimulating for thought about the reading that I have adopted this general plan for all the chapters in this volume. C. E. S.

Chapter III

MUSIC BETWEEN THE AGES OF SIX AND TEN

The question "*When* should music education begin?" is now coming to be "*How* should music education begin?"; because we now recognize that music should play a large role in the first five years of child life.

Soon after six the child enters school. Here the principles of educational psychology, now so effectively applied to other primary and elementary school subjects, have revolutionized the presentation of music. This has of course been favored by the recognition of music on the level of the three R's. Music has come to function in the school not only as something to be learned but primarily as something to be lived. Primary teachers are, or will be, trained specifically for this subject. The old conflict between enthusiasts for rote singing, on the one hand, and for technical sight reading, on the other, is vanishing. The approach to music is following new avenues involving diversified action, creative imagination and thinking in music, recognition of individual differences, freedom for individual expression of musical feeling, opportunity for sampling various avenues of choice in expression, the association of music with play, dance, and dramatic action, opportunity for hearing music at the child's level, avoidance of the fostering of a narrow precocity, and recognition that there is music everywhere—in speech, in play, in nature.

Parents who now aim to provide private lessons for formal training in some aspect of music must lay their plans in the light of all these facts which have come into view so strikingly in the school. They must understand and evaluate the significance of this movement in the school, the new status of music, the new role of private instruction in music, and the availability of a private teacher who can dovetail with these new facilities and responsibilities. In the hope of giving some helpful suggestions in regard to this planning, I wish to present some psychological

considerations which are, at the present time, reconstructing the theory and practice of private lessons in music at this age.

A broadened conception of music. In Chapter II I pointed out that children coming out of a favorable music situation in the home, the preschool, the kindergarten, and other school and playground activities have attained rather astonishing achievement. Let me repeat, for emphasis:

"During the first six years there should be no formal musical instruction; but by the end of that period, the musical child should have gradually acquired a sense of appreciation for musical sounds, pleasure in self-expression in musical intonations, confidence in his ability to compose a tune, some proficiency in singing and good speech, and some degree of satisfaction in free playing with an instrument."

This achievement now accomplished in many preschool communities presents a challenge to the primary teacher and the supervisor of music in the public schools. And, where pupils come without such preparation or background, they must begin from scratch and offer a substitute for it in more concentrated form.

Private lessons should be built upon this background and designed to carry this type of program forward during the next four years with progressive enrichment of opportunities on the basis of talent thus revealed and with these types of activity as a goal in the beginnings of formal training.

This point of view turns a large part of the job of the private music teacher over to the primary school where it is favorably developed; because only in the group activity and in the avocational attitude with the avocational atmosphere under technically qualified teachers can this program of musical education find its best fruitage for children in general. The private teacher is falling from her high pedestal of the power to cast the child's musical mind in the pattern of her own image within a limited musical skill at this age. On the other hand, the ideal of the school situation which I have pictured will play happily into her hands by furnishing a background for a systematic study of voice or a particular instrument.

In this procedure, the child's interest may well concentrate around a single instrument. Furthermore, we are just awakening to the fact that voice development should begin in this period. New techniques for the development of a beautiful voice are coming in, and are adapted to this age.

They should play an important role in this orientation period, and the child's interest may concentrate around this, as well as around an instrument, especially through the association of artistic forms of speech with song.

The analysis of talent. Music educators are coming now to a recognition of the principle that musical education, public and private, should be given in proportion to the possession of natural talent and in the direction for which the most favorable indication is found on the ground of specific talents. Such talents reveal themselves through the daily activities of the schoolroom, where the alert teacher understands their significance and directs them wisely. There are certain basic abilities which are favorable to a musical life. Some of these can be measured accurately with the *Seashore Measures of Musical Talents* in a revised edition now available through the RCA Victor agencies. They are concerned with the senses of pitch, loudness, time, timbre, and rhythm, together with tonal memory. These may be given individually to children between the ages of six and ten if administered with good judgment and the child is not required to write the answers. However, if the teacher is trained in the musical analysis of talent and the critical observation of children's behavior, inceptive achievement, and interests in the musical situation, it is possible to proceed without instruments, since the teacher will know what to listen for and will be competent to observe with sufficient accuracy for the purpose in hand. After the fourth grade, "measures of talents" may be given either as individual or as group tests. They may furnish a key to the most natural development of musical type; such as, the tonal, the dynamic, the temporal, or the balanced types, on the basis of degree of responsiveness to each of these factors as well as on other specific adaptations for voice or instrument. The teacher who has such concrete facts in hand will observe and plan the child's development more critically. Every private teacher should have a conception of the significance of individual differences in talent, and might well utilize measurements of this kind in the studio. The progressive private teacher should also be able to serve parents in an advisory capacity on such matters.

It is not necessary that the teacher should be an expert in testing; but it is essential that she should understand the nature of musical talents and have the sort of insight into child life which has been developed so splendidly in recent scientific studies of child behavior, especially with reference to the recognition of natural abilities, the early fostering of these, the devising of

effective motivation, the awarding of praise and blame, and the setting up of standards of achievement recognizable by the child. It is, furthermore, reasonable to demand that the teacher herself should present a good voice in song and speech, have some proficiency and ingenuity in the manipulation of simple instruments, and feel deeply the love of music.

Group instruction. The old-fashioned piano lesson was at fault in various respects. First, time was taken to tell the pupils, individually, the elements of musical notation; second, the main function seemed to be to make the pupil "get" his lesson; and third, the child was expected to acquire a love of music through the technical approach. These things are now changing. The pupil now coming from the respectable primary school already has acquired the elements of sight reading, both from ear to eye and eye to ear. Any new element in notation can certainly be picked up incidentally as needed, without waste of time. The position of the teacher as a taskmaster is also disappearing. True, that eliminates many a pupil, but probably without much loss to the musical life of the community. The child comes to the private teacher already motivated with a feeling and urge for music. Musical achievement is no longer counted in terms of the number of lessons taken.

The function of the teacher is far more to motivate than to teach. The thing that counts is how long and faithfully the pupil works to acquire the proficiency which can only come through practice. What can be taught in one lesson should suffice for several times as much practice as is ordinarily expected. Better than having the teacher crack the whip for many periods would be to give the time available for lessons to the pupil for practice and use more remunerative rewards, even a percentage of the teacher's fee, to encourage self-help in a larger assignment through adequate practice.

One of the significant advancements in private instruction for children of this age lies in the direction of class or group instruction. No effort is made to force all children into the same cast. Small groups are formed on the pattern of chamber music, taking children of matched abilities and interests and using music progressively adapted to their level. Many forms of class instruction have failed on the ground of the teacher's inability to use informal procedures in some form of project method. Duets, trios, quartets, all in the competitive and play mood, can accomplish great things, even with the youngest children. Few private teachers have awakened to a realization of the fine possibilities in that approach. This method is

especially adapted to primary instruction in private music schools in which there are enough pupils to make competitive promotions from group to group. Most of the musical information and the motivated drill can be accomplished through the group. The recognition of this principle may lead to the development of extracurricular and private organizations under an inspiring teacher or group of teachers for private instruction.

Formal lessons delayed. The principal point I wish to stress is that musical education should not begin with formal lessons on one instrument. Except in the case of rarely-gifted children, such specialized instruction should come naturally after the general musical interests have been awakened and the natural abilities have been revealed. Technical private lessons for children in general should therefore be begun considerably later than has been customary. In this there is a threefold saving: First, except in rare cases, rigid technique of instruction can be responded to much more economically after the age of eight or ten than before. A ten-year-old will acquire more than twice as much in a single lesson as a six-year-old. This will apply even to the much "touted" necessity for early finger development. Second, during this period the child should have the freedom to try himself out spontaneously with diversified encouragement in the development of specific interests; and third, it will take the aspect of drudgery away from the music lesson. To these may be added the fact that this liberal procedure helps to give the child a feeling that he is living music rather than learning it. It involves the play attitude in the acquisition of an art. The attitude of feeling the necessity for hard work, which is a very real necessity in music, can best be cultivated formally after the age of eight or ten.

A sympathetic listener. Although circumstances may alter cases, facility in piano playing might well be regarded as a foundation work in the approach to other instruments. The child's preference for a particular instrument is generally childish and will change in the normal course of development. An analysis of case histories would make an interesting study on this point. The development of skills in a particular instrument should always be accompanied by opportunities for sharing the pleasure in this skill with other children and the presence of a sympathetic listener in the teacher and the parents. What the child of the primary-school age needs is more a sympathetic and critical listener than a task-driver. This type of approach will discredit the now so prevalent artificial ways of symbolizing

music by attempting to force the teacher's affected and stilted imagery upon the child's musical mind, which may run naturally in much more effective channels. It is a notable fact that the great musicians who emerged as very precocious made their early and distinctive progress far more through freedom for self-expression than through instruction from the masters.

In brief, the private lesson to the child should pattern, at the child's age level, after the procedure followed in adult instruction in music at its best; namely, that of sympathetic and inspiring criticism and guidance rather than the dealing out of predigested pellets of interpretation and technique. The general attitude to be cultivated in the child should therefore be freedom for self-expression rather than mere willingness to absorb set tasks. In this attitude the problem of scales, exercises, and calisthenic techniques will come in the natural course of events without being forced.

Music lovers vs. virtuosi. One of the first essential steps in training at this level is to educate the mother to the notion that only in very rare cases will the child become a virtuoso or a professional musician, and that she has no right to pose as an exhibitor. The normal child is richly endowed with powers for diversified development. By too early emphasis upon a "gift" it is possible to produce monstrosities and pathological temperament. True, a gifted child beginning, for example, by taking lessons at five and being effectively motivated, may produce extraordinary results before ten, but generally at the expense of a normal development of the child as a person. An ill-guided enthusiast can make of the bright normal child a mathematical freak or a contortionist before the age of ten; but who wants that for the welfare of the child? It is a form of human sacrifice. The time for intensive specialization should normally come after the age of ten. Parents and teachers should shun the development of precocity as they shun disease. Indeed, in ninety cases out of a hundred, excessive precocity is a form of disease, a distortion of the normal personality.

The goal in musical education in this period should be to recognize individual differences, natural capacities, and native interests and urges in their natural stages for the development of a well-rounded personality. We have long since abandoned the notion that every girl should play the piano. What we need to learn now is that we should not allow the musically-gifted child to die with all the music in him, and that the musically gifted should not be exploited. The middle ground is that all children who have musical

ability should learn to love music and live it naturally, each according to his ability.

THOUGHT REVIEW

General Principles

- (1) Music should function early in the life of the child.
- (2) The primary grades should constitute a period of orientation.
- (3) By the end of this period talent should have been discovered and analyzed.
- (4) Voice training should be given a permanent place in all the grades for both boys and girls.
- (5) Music is a skill which should be learned and lived rather than be taught.
- (6) Wise motivation should replace much teaching.
- (7) Nomenclature, sight reading, and musical information should be taught in classes, not by individual instruction.
- (8) The principle of rivalry should be encouraged.
- (9) For most children formal music lessons may be begun most effectively about the age of ten.
- (10) Forced precocity and exhibitionism should be discouraged.

Questions to Consider

- (1) Why is development of the love of music the principal objective at this age?
- (2) What voice training, aside from rote singing and speech, should be offered in the grades?
- (3) Should the piano be favored as the first instrument?
- (4) Is it possible to have a primary-grade "orchestra"?

- (5) What is the function of achievement tests in music at this age?
- (6) What factors have been most effective in putting love of music into the public schools?
- (7) How can progress in music be enhanced by children's opportunities for performing among themselves?
- (8) Should any children be excused from participation in musical instruction offered in the primary grades?

Discuss These Situations

- (1) Here is a gifted young woman, well trained in music and in psychological and educational methods, zealously devoted to her art, who desires to establish a career as a private music teacher for children. Characterize her procedure in contrast to the procedure of her professional forbears.
- (2) Here is a young man or woman, similarly gifted, and well trained and critical. What pitfalls of prevalent tendencies in public-school music of today should he or she avoid?

Chapter IV

MUSIC AND YOUTH

There is a distinctively musical period in life: the period of youth. Youth is the age of emotional response and of social awakening, the age of serious play, the age of decision and elimination, the dominant learning period, and the age of freedom and leisure. Before this period the life of the child has been relatively tranquil. After this period the occupational affairs of life are more exacting. Before this period most children participate in music in a routine way without professing it. After this period large numbers of those who have had training or have expressed their enthusiasms in rich participation in the musical life cool off, as it were, and continue their musical activities in a more or less perfunctory way, except for the few who have taken up the art professionally or continue as enthusiastic amateurs.

YOUTH, THE AGE OF MUSIC

The emotional age. For the present purpose we may think of youth as represented by the age of the teens, usually beginning in the high school. It is a brief period of storm and stress, emotional awakening and emotional struggles, in which the various emotional drives, more or less latent before, assert themselves, often to cool off or to be attenuated in later life. The older psychologists spoke of it as a period of rebirth, the passing from the period of protected and directed life into the emergence of a self-asserting personality. At the time of this awakening, life is largely emotional; it is the period of ardent love, of social awakening, of sports and play of all kinds, of fortes and faults, of artistic efforts, of devotion to ideals, of awakening and emergence of controlled imagination, of the development of the great enthusiasms for life. Poetry, conversation, dancing, heroic stunts, emotional adventures of all kinds have their fling. This emotional tendency finds its expression in the language of emotion, of which music is one.

The age of serious play. Youth is the age of serious play. Before this period the child has spontaneously played much and hard enough, but now youth enters into specialized sports, finds a play outlet in one or more of the arts, and with the play attitude takes up serious roles in social responsibilities. All these are comparatively new and are, therefore, entered on with fresh enthusiasm. Emotion has a way of burning itself up. Toward the end of the college course, youth, having had its fling, feels fed up on sports and after that may not even sit on the bleachers or the side lines. The same is true of many of the artistic outlets. When youth comes of age, the ventures into poetry, music, dramatics, and various other forms of art cool off for the majority. This loss of interest is especially noticeable in the social activities, in which the vocal arts, the dance, and other instruments of wooing play so large a role. It should, however, be said that the better our education, home life, and vocational activities are organized for right living, the less is the break at the beginning of this period of youth, as well as at the end.

The age of decision and eliminations. The persons exceptionally gifted in music are precocious in that their artistic awakening comes earlier, and yet the real assertion and the serious functioning of their musical inspirations

appear most characteristically and pass through their crucial period in the teens, after which large numbers of those who have gone through the fire of struggle for survival and mastery pass out of the picture.

The educational age. Youth is the dominant educational period. Those who in this period qualify for higher education can and should continue their studies with organized effort and with confidence in success; but, on account of the American popular demand for higher and higher education, many of those who should be eliminated during this period float on, ill-motivated, ill-directed, floundering through college or professional education without regard to the worth-whileness of the procedure. Unfortunately American education has not made adequate provision for the diversifying of training, especially in the practical outlets at this age.

This educational dominance of the period of the teens is particularly true for music. Unfortunately, large numbers of youth naturally endowed with the power of enjoying music have not even been discovered up to this time, either by themselves or by others, and they may pass through this last stage of opportunity without being discovered and often without realizing what they are missing. At this age the prospective musician will begin to try himself out, will discover the necessity for untiring and well-directed work, will concentrate in the field of his most natural musical outlet, and will work hard in the face of the imminence of the decision as to success or failure in his own judgment, in the judgment of his teachers, and in the judgment of his companions and public. If successful here, he may continue his musical education in specialization, since music is essentially a body of skills. At the same time he may show a due regard for the necessity of a broadening education under larger horizons. Fortunate are they who follow these sound procedures.

However, many students overlook the fact that, if their music is to be an avocation, it is to be pursued merely for the pleasure inherent in it. They may struggle along simply because they have had initial training in this field, but without adequate motivation or anticipation of significant results. Achievement in music is an artistic performance, and we should not add recklessly to the large group in society who feel that their early efforts were wasted and who have to go through life apologizing for their failure or mediocrity in the artistic sense. At the same time we must bear in mind that satisfaction in musical performance for the great majority of persons comes from the pursuit of the cruder forms of music, especially singing. Group

singing, ordinary piano accompaniment for singing, dance music, and ragtime of all sorts should not be discouraged in so far as they form a natural outlet for those who pursue them. After all, "high-brow" music plays a very small vital role even in high society.

Unfortunately, as in academic subjects, the real elimination of the unfit is not accomplished effectively, and there are masses of music students dribbling along with at best a low mediocrity as the destination in sight. Since music, to be of service in later life, must be either professional or eminently satisfying as an avocation after advanced work in the subject, the necessity of making discriminating and wise eliminations before a youth is of age cannot be stressed too strongly. A wise selection and guidance at this stage should, however, in no way discourage those who get a genuine satisfaction out of music and are wisely motivated, especially for the keeping of an avocational interest alive with a modest degree of attainment.

We must face the fact that most of those who are highly gifted in music and are deeply devoted to it will not enter the profession of music and may be engaged in such exacting occupations as to limit the time that they can devote, avocationally, to their favorite art. Yet of all the lovers of music, these are perhaps the ones who get the greatest enjoyment out of music and contribute most happily to the musical life of the community.

It is not sufficient to get children started in musical education. They must also be motivated and given the outlets for self-expression which tend to have lasting value in their lives. A drab picture is drawn by comparing the number of those who, through high school, have had the privilege of good training with the number of those in the best social groups of our cities today who actually contribute, and take pleasure in contributing, to the musical life of the community and to their own personal moments of leisure. Here is a challenge to adult education: a challenge to find and encourage self-expression in music in adult life. Nevertheless, a strong argument can be made for musical education even if it functions only during the period of adolescence.

The age of leisure. Youth is, in a sense, the age of leisure. We think of a gentleman of leisure as a person who, by virtue of his being a gentleman, is active in pursuits which are not necessary for the earning of his bread and butter. If he is not active, he is not a gentleman. If his activity does not add to a genuine enrichment of his own life and the life of the community, he is not a gentleman. If he does not know how to play, he is not a gentleman.

The same ideas apply in principle to the concept of a lady of leisure. The adolescent youth has many of the characteristics and the opportunities of a gentleman of leisure. These opportunities do not mean an easy life, but a life characterized especially by the freedom to do just the things that he likes to do. These opportunities include, for many happily adjusted personalities, the development of the art of loafing: the substitution for mere sitting, for mere gossiping, for mere eating and drinking, and for degrading pastimes, an active life of restful, satisfying, wholesome self-expression in the play attitude.

The high-school or beginning college student is not, as a rule, working for a livelihood, is not held down to the specific demands of a trade, a business, or a profession. He is in a period of preliminary skirmishes, and the public school programs provide liberally for these. He may spend his leisure time in the strenuous work of athletics or of any intellectual, artistic, or social pursuit lying outside the demands of the curriculum; he has time for such excursions.

One of the happiest ways of spending his leisure time may be in the enthusiastic pursuit of musical activities. When he gets into college, these opportunities are more restricted; when he gets into his job, they are still more limited. Whatever the line of pursuit, these extracurricular activities are indulged in as a form of play. Nobody, however, can express in play all his intellectual, artistic, and social dreams or urges. Therefore the gentleman of leisure in the high school is forced to make a selection, and, in doing so, he rightly follows his natural bent of mind. If he casts his lot with music, he will train hard, and have the social satisfaction of supporting his musical group. Music becomes a game, involving contests with defeats and victories. It becomes a topic for conversation in social leisure. While the high-school program and the junior-college program do not emphasize rigorous technical training as a part of the academic curriculum, the majority of students get their musical satisfaction from the way in which they utilize their leisure for music. It is significant that at this age the stress is not on the hearing of music but rather on participation in musical performance. This period is the age of action, perhaps the best type of preparation for the listening stage since it follows the educational principle of learning by doing.

MUSIC FOR YOUTH

During the present century most extraordinary progress has been made in the provision of musical facilities for youth by the recognition of music as an academic subject, by the early and vital training in the grades, by the development of group activities in voice and instrument as a dominant extracurricular activity, by the motivation of training through opportunities for public performance and contest, and by the popularization of music through phonograph and radio.

Up to the end of the past century youth had had no significant opportunities for music in the scheme of things. Music had to come through the taking of private lessons, often a drudgery without regard to likes or dislikes. The approach was purely technical. The mastery of the scales, though technically significant, was not inspiring to the emergent musical mind and did not reveal the vast vistas of opportunity for self-expression in music. Music lessons were limited to those who could afford to pay for them. Few people had heard any good music. It had no significant status in the program of public education. Youth had no chance. *Music for youth has been discovered in the past forty years.*

Music, an academic subject. When music gained recognition on a par with the three R's in primary and elementary education, America entered on a new era. National music organizations of teachers developed. These began with the slogan, "Music for every child at public expense." Realizing the futility of making every child musical, they modified this slogan to, "Music at public expense for every child in proportion to his natural ability." The market was flooded with educational music books and systems. Time for musical training was set aside in the regular curriculum of the schools. It was recognized that only a person who has musical ability and training can teach music effectively. Scientific principles from other subjects were adopted. Specialized training for musical supervisors was provided in teachers' colleges. Teaching was improved. Groups of children were segregated into vocal and instrumental organizations, both for curricular and extracurricular activities. Instruments were furnished at public expense. Credit for music gained current coinage on a par with history and chemistry. The introduction of principles of educational psychology led to selective

admission and elimination and gave the teachers a basis for the administration of praise or blame in achievement. Few public school subjects can point to a similar epoch-making stride. These achievements today give youth a chance in music.

Orientation in the grades. For the musical life of youth at and soon after the high-school age, the training in the grades was most significant, and that, in turn, was rooted in a national awakening to the possibilities and the responsibilities of music in the earliest years in the home and the kindergarten. This awakening was strengthened by the recognition given art in general. Music early became associated with the speech arts, the dance, and other games. The development in the graphic and the plastic arts, though slow, was somewhat parallel with music for these early years. In the eight years in the grades there developed a process of selective opportunities for the pursuit of music so that freshmen entered the high school as a group fairly differentiated on the basis of abilities, opportunities, likes and dislikes, and ambitions for music.

This differentiation had a twofold effect on the high school: First, the elimination of pupils who, on various legitimate grounds, did not qualify for musical training at public expense at this stage and, second, the beginning of specialization for those who had previously found themselves or who at this stage made a happy decision. Thus the development of music in the high school has followed logically on the early development of music in the grades, and in like manner the development of music at the college level is now beginning to follow the development of music in the high school.

Group activities in voice and instrument. Youth is the age of learning by doing. This principle has been implemented in high-school education by the development of group activities, both in theoretical and in practical music, both in the curricular and in the extracurricular activities. Throughout, the emphasis has been on action. Instruments are supplied at public expense; opportunities are given for participation in programs of entertainment; and a stimulus has been climaxed by the introduction of local, state, and national contests for groups and for individuals.

The adoption of the principle of group instruction at this level has had many significant advantages. It draws large numbers of pupils into group activities for the social value of the activity in itself and for the social value of participation in public life. It has lessened the cost of musical education many times over by training ten or a hundred pupils in a group as

effectively, in many respects, as they could be trained by individual instruction. Group instruction has, perhaps, been the strongest leverage for the motivation of musical pursuits in the school. It has increased the pursuit of individual instruction on the basis of discovered needs, for example, in the mastery of a particular instrument for participation in the band or orchestra or in the development of a voice discovered in the group. The individual instruction has thus been motivated and made to fit into particular niches in the musical groups or in the choice of vocational pursuits in society.

Contests. The development of contests both in individual and in group performance has had a profound effect on the development of music in youth and is destined to serve in large part as a clearinghouse. Educators recognize many drawbacks in the plan of contests. Preparation for the contests requires excessive time and highly trained teachers. A contest becomes an occasion for the revelation and the objectifying of differences in individuals and groups. It involves some little expense. It tends to discourage the pupils who cannot qualify.

But let us look at the other side. In the state of Iowa, for example, the state-wide contests, which have been conducted for about the past ten years, have done far more than anything else to vitalize music education in the high school. Four years ago I became acquainted with a group of ten girls who came from a little town three hundred miles away. They had not placed in the contest; they had had no chance. I supposed that this contest would be the last that we should see of them, but the next year they appeared in a body—happy and optimistic, in a fighting mood—and won honors. I asked how they accounted for their success, and they said, "When we got home and reported that we had had no chance in the state contest, the commercial club of our town raised a purse and employed a teacher to come into the high school and train us in music." When the commercial club goes into its pockets in the interest of putting music on the map in its community, music is beginning to function in that community.

Nobody cares to tune a fiddle for four years; but, if he has the opportunity to play a tune, he will gladly keep the fiddle tuned. This fact has been shown on a large scale as a result of the contest plans. Not only have the final local, district, and state contests become goals, but every daily practice has gained an objective and is a topic for social conversation. The contests have made each participant critical of his own performance

and the performance of the group and have also made each ambitious for improvement. They have raised standards of selection in musical material. They have resulted in the giving of higher musical education to large numbers of gifted persons whose talents would not otherwise have come to the serious attention of parents or educational interests. The methods of judging these contests have been greatly improved in the direction of offering encouragement where encouragement is due without discouraging more modest attainments where they are indicated.

Educators used to think that football was the only thing that would arouse enthusiastic support in the field of avocational activities for youth; but, as I have witnessed the attitude of high-school pupils in training for musical contests, heard reports of the attitudes of the parents and the backers, and watched the culmination of enthusiasm at the annual contest, I can say that there can be, for youth, as substantial enthusiasm in contests of music as there is in football. In other words, the youth of the state have entered into music in the spirit of play, with determination to train well, play hard, and win if possible. Enthusiasms for a center of interest in public and in private have been found in this music.

The hearing of music. Within the past forty years, also, America has for the first time begun to hear music, good, bad, and indifferent. When the first phonograph came to our city, the host at an evening reception furnished music in a mysterious way. He had installed the phonograph in a niche curtained off in the hallway so that the direction from which the sound emerged was difficult to detect. He played three or four songs by a male quartet, which were richly applauded by his audience on the supposition that he had stationed a male quartet somewhere in the house. Music had come into the social group in a mysterious way.

Throughout the foregoing discussion I have stressed musical performance because action is the way of youth, the way of education, and the way of any preparation for the listening mood and critical attitude. The development of the opportunities for listening has, of course, been unparalleled in the history of the world. The masses are given the opportunity of hearing good music. Many have taken the attitude of sneering at the music offered by phonograph and radio on the grounds of quality of rendition and of choice of type; but it is safe to say that what is offered is what people want, and there is some basis for saying that to a great extent what people really want is what is good for them. On the other

hand, the furnishing of music at all levels has unquestionably had the marked effect of raising the American level of musical appreciation step by step. People can be educated to listen, and in a marvelous way the radio has really taught people to listen. You cannot learn to swim without going near the water. Radio has had the effect of discouraging some persons from the development of musical skills in performance, because they realize that what they can hear is so superior to anything that they can do. On the other hand, radio has stirred the musical interests of the masses into appreciation of the possibilities of music at their respective levels and not only has made them whistle the tunes but has encouraged them to participate in various modes of performance. Listening to the "canned" music has certainly gone far to develop the ability for listening to legitimate music on the stage.

Some years ago I had the privilege of being the guest of the city of Copenhagen for a week. Among the entertainments offered by the city was a grand opera. The invitation came in the form of a large, beautifully engraved card of marked distinction and good taste. Attached to it was the actual ticket to one of the best seats in the house, and the price printed on this ticket was, in American money, twenty-three cents. It can be said almost literally that, when grand opera is played in Copenhagen, people of all classes come out and tend to listen with appreciation and more or less discrimination. The people have been educated to hear music. It has been brought down to their level. Therefore music functions in the social and home life of that city. Music of a very high order furnishes the entertainment in the parks and the public halls. That tradition is fast coming to this country, and the effective training of our youth will do much to hasten the day.

THOUGHT REVIEW

General Principles

- (1) The high-school age is the most significant period for the development of an avocation.
- (2) With the coming time for leisure, education for avocation is a sound educational policy.
- (3) It is fortunate that musical education at this age is elective.
- (4) Education for musical taste must be gradual, not by leaps.
- (5) Expansion of high-school music may advantageously be extracurricular in large part.
- (6) Music contests have greatly aided in the promotion of musical education.
- (7) The radio and the phonograph have in recent years been the most effective media for the dissemination of musical information.
- (8) It is quite natural and well that the fervor for music should cool off for many at the end of the teens.

Questions to Consider

- (1) Should the schools supply musical instruments at public expense?
- (2) If so, what selective procedures and restrictions should be set up?
- (3) Is the author's conception of the characteristics of adolescence generally accepted by educators?
- (4) What are the evils of music contests?
- (5) What are some of the unfavorable musical effects of the radio?
- (6) What problems arise in connection with the change of voice in boys?

- (7) Should the high-school *a cappella* choir exhibit the vibrato in singing? (See the author's *The Vibrato in Voice and Instrument*, University of Iowa Press.)
- (8) How can high-school musicians render larger religious, social, and patriotic services in the community?

Discuss These Situations

- (1) Last fall (1939) I saw seventy-eight marching high-school bands with over 6000 instruments in a two-mile parade on the morning before a football game at the University of Oklahoma. Here was a cross-section of the youth of the state, many of them Indians; all in uniform in dancing march for musical display. Consider pro and con the implications of this phenomenal outbreak of music from (a) the musical, (b) the social, (c) the educational, and (d) the economical points of view.
- (2) There is a splendid tendency at present to integrate the school with social life, community services, and part-time employment. What features in such integration seem most promising for music?

Chapter V

THE MUSICAL TEMPERAMENT

What makes a musician temperamental? Tell a musician he is temperamental, and he will take offense. Yet perhaps the thing in his personality of which he is most proud is the possession of a musical temperament. This characteristic inconsistency has a basis in psychological fact; namely, that the exhibition of artistic temperament frequently leads to attitudes and actions which the rest of the world may criticize and view with amusement; but, on the other hand, the finest expressions of musicianship would perhaps be impossible without the possession of an artistic temperament.

Many persons who pass as musicians are neither temperamental nor musical. A great many of those who ply the art of music do not have musical minds in any basic sense. Their art consists in certain skills often built into a purely matter-of-fact organism. I therefore see no reason why such people in the musical world who do not show any artistic temperament have any reason to boast of the fact.

The highly temperamental musician is a species of genius in some degree. As such, he has been described by musicians, scientists, and psychiatrists in a very copious body of literature on the subject, especially that dating since the time of Lombroso, who regarded all geniuses as degenerates. The modern psychiatrist has frequently presented a picturesque view of the temperamental person in terms of the rising science and art of psychiatry, which is supposed to explain all deviations from normal behavior in terms of psychopathology. The old saying was that we are all more or less sane; the psychiatrist today says we are all more or less psychopathic. The current literature on mental hygiene is characterized by this point of view. Another point of view has proved fascinating since the coming in of Freud, in that deviations of this sort are accounted for in terms of suppressions, defense reactions, and other manifestations of the libido.

Much light is thrown on the problem by the great musician's exaltation of the artistic mind in action, thus revealing essential temperamental traits on the basis of first-hand experience and artistic theory. A temperamental musician is the most merciless portrayer of his own species. Since the temperamental person is always an interesting person, all of these accounts are lively and each revealing from its own point of view, giving us much genuine insight into the nature of the temperamental person in all fields.

Descriptions of the musical temperament are, as a rule, vivacious and luminous, thrillingly interesting. If a competent analyst would follow Stokowski for one day during the season and picture in high lights, full of concrete examples, the experience and behavior of this great conductor from the whirl of emotional enthusiasms and conflicts into which he awakens in the morning to the convivial abandon during refreshments after the evening performance, we should have a spicy and animated picture of the musical temperament. If written by the smart commentator in the columns of *Esquire*, it might even be racy. Such pictures have often been drawn in fragments. A similar picture might be made exclusively by direct quotation from biographies, autobiographies, and letters of great musicians, such as Beethoven and Wagner, whose lives are now on record. The temperamental musician is not competent to analyze himself because he does not see himself as others see him. Literary chats about musicians are full of quirks and eccentricities giving realistic examples of a temperamental life. These might be organized into a fairly complete characterization of a distinctive musical temperament.

All such exhibits would be of the emotional type, dramatic extravaganzas, interesting and significant, and there are many such extant. The inceptive science of the psychology of music takes a different point of view, aiming to account for the highly temperamental person partly in terms of heredity and partly in terms of inherent elements in the situation in which he lives and performs. The music psychologist must content himself to deal in a cold analytical way with verifiable and orderly facts in order to contribute something toward a scientific foundation for a systematic and functional psychology of the musical mind. In attempting this approach, I realize how tame and stilted such an effort must seem to the fiery musical temperament; and even at its best the psychologist will regard the account as speculative, because it is seldom based upon experiment. I have delved into musical biography and autobiography of great musicians with an eye

toward the discovery of their outstanding mental characteristics from a psychological point of view. On this basis I must make a bold venture by attempting to trace the elements in a musical situation which lead to the development of temperamental behavior and thus contribute toward the answer to the question, "What makes a musician temperamental?"

Physiological irritability. The highly-gifted musician responds physiologically to sound stimuli to a very high degree because he has inherited a genetic constitution which is anatomically and physiologically exceptionally responsive to sound. In other words, quite apart from consciousness of sound or thought of music, his physical organism responds to acoustic stimuli of all kinds which keep nerve and muscle in a state of tension. This tends to create a state of unrest and irritability. Without leading to actual hearing, it may arouse associations of a sort of dreamlike or dramatic nature which may play a very large role in the conscious life. It may create a state of well-being and happy associations, but perhaps more frequently irritation and noxious day-dreaming associations and emotional eruptions. The sounds may come from a squeaking chair, the sizzling of a kettle, the song of a bird, the cry of an infant. Most frequently sounds affecting the organism in this way are inconspicuous and commonplace in the environment; but they may often be strong; such as, the rattling of a train or the chattering in a crowd, of which the musician does not become conscious, although physiologically irritated.

Tonal sensitivity. All great musicians are highly sensitive to sound in all its elements. A physiological irritability acoustically leads to a profuse awareness of sounds. This sensitive and selective hearing gives the musician a richness of material in the musical medium. He becomes intolerant and rebellious to disturbing sound stimuli which have no such effect upon unmusical persons. To him, the world of sounds has infinite richness of resources for musical pleasure, but equally exceptional resources for the suffering of musical pain.

Artistic license. To the trained musical individual sounds are heard as different from what a matter-of-fact listener hears. The hearing of pitch, loudness, time, and timbre is not in the ratio of 1:1 with the physical sound, but always runs into artistic analysis and interpretation with artistic license. The musical interval, the dynamic phrasing, the rhythm, and the tone quality are always heard in relation to their artistic setting. Here we find auditory illusions operating in their fullest glory. The pitch value varies with

the quality of the sound. Time may be a substitute for stress and vice versa. A subjective rhythm is richer and far more realistic than the physical rhythm. The quality of tone is heard in relation to its musical meaning. To the musician, the hearing is not so much a question of true pitch, formal accent, temporal rhythm, or vowel quality as it is a matter of musical balance and a recognition of artistic deviation from the true. Meticulously exact performance of a Bach score would be musically intolerable. Notes are frail symbols. The performer must interpret even the shortest measure rhythm or single note value. Thus, while fine sensory discrimination in all the aspects of sound is essential for correct hearing and tone production, the more essential thing is his ability to play with artistic power and impulse in hearing and producing artistic balance and artistic deviation from the rigid. In this artistic balance and deviation, he may be guided by certain artistic rules, but his direct emotional interpretation is far more significant. In this lies individuality. He is constantly tempted to be extravagant.

Thus, in all the variants, combinations, and modulations of pitch, loudness, time, and timbre, the musician hears, feels, and gives meaning to fine and subtle distinctions, many of them quite divergent from the physical tones. At this level, temperament shows itself in exceedingly fine responsiveness to tones which may be a matter of utter indifference or impossibility to the unmusical. This capacity is largely inborn, both in the way of sensitivity to sound and a general nervous, if not neurotic, disposition, and is in itself enough to make the musician different from other people. Artistic license as a medium for self-expression is, therefore, clear evidence of a musical temperament.

Ear-mindedness. The successful musician is ear-minded as distinguished from the painter who is fundamentally eye-minded. This ear-mindedness grows out of his genetic auditory constitution at birth and develops through the practical use that he makes of the various attributes of sounds. Fundamentally, it is the tonal image which, in the great musicians, is practically as vivid, stable, and complete as the ordinary perception of the actual physical sound. His memories, his imagination, his creative work practically all operate in terms of his powerful auditory image, usually supplemented by strong motor and visual imagery. This makes him different from the businessman, the objective scientist, and the man on the street. The fact that he can live subjectively in this tonal world gives him a type of isolation in which he feels the superiority of his power and becomes

disposed to assert his rights, privileges, and dominance in a domain in large part separated from ordinary affairs. Thus, the powerful imagery becomes one of his richest and most fundamental resources for exclusiveness as a musician and, when cultivated in the field of artistry, tends to set him apart from the rest of the world.

Affective response. Since the business of the musician is to hear and produce beautiful effects in sound, he differs from the ordinary listener in that sounds of all kinds not only have intellectual orientation value but are habitually responded to in terms of beauty or ugliness, pleasure or pain. They give him warmth or chills, both of which create feelings of unrest. Instead of identifying a sound as the rumbling of a train, he identifies it as ugly—something which disturbs and irritates him. Instead of identifying the sound as the song of a bird, he responds vigorously in terms of likes and dislikes.

Emotional pursuits are usually sexually stimulating, and persistent emotionality is likely to manifest itself in love "scenes", good or bad. As a mate, the temperamental musician may be a most ardent and exquisite lover or a most irritable person to live with. This is aggravated by the fact that the artistic behavior is worshipped by the opposite sex, often to an annoying degree. Hence, love, self-defense in love, and a sense of superiority in love are keynotes to the temperamental musician's life.

The esthetic mood. The musician is in search of the beautiful and therefore, conversely, responds unfavorably to the ugly. His professional life is, in the main, emotional as distinguished from the intellectual life of the scholar in other fields or the action-patterns of men of affairs. Whether he is a virtuoso, a creator of music, or a director, he is working on emotions through emotions, trying to re-create for the listener the feelings with which he himself is imbued. He lives so intensely and habitually in this activity that he becomes recognized as highly and persistently emotional. This extreme emotionality in his daily work sets him off against the matter-of-fact mind. We say of the intensely artistic person in action that he burns himself up. The emotional life is expensive and flitting; it flashes and explodes and is in danger of running out of control.

This emotionality tends to transfer not only to other forms of art but to ordinary things and situations; such as, money, raiment, or social amenities. Sometimes this takes the form of the characteristic bohemian. He may spend his wages on payday and starve the rest of the month in utter

complacency. All his life tends to be set at high tension. He lives dominantly in a mood and therefore often becomes objectionally moody through his impulsive behavior.

Exhibitionism. There is an accretion to the musical temperament in a sort of hierarchy of defense reactions which may be characterized as exhibitionism. The musical mind is pulling on one end of a leash, as it were, trying to drag the more or less resistant and incapable into his own beautiful emotional life, and he feels the drag. Therefore, he becomes impatient and uses ways and means of exclusiveness in withdrawal from the world, or he takes the opposite attack—display. To him, countless means of personal display justify their end, the glorifying of his noble art. Therefore, we see the musical temperament in this artistic form in the manner of living, eating, dressing, and sleeping, and in the demand for hero worship. We see this in the extreme form in the conductor who feels that he must have his choir, his orchestra, his band members each individually at his command. He is like a general in action. For this purpose, he must pose as a great authority, as a hero standing for something superior in the way of personal interpretation, as a critical judge of the beautiful, as having undisputed power, feeling the necessity of imposing his own emotional individuality on a comparatively cool and often resistant group. This is true on a smaller scale in all musical leadership, composition, performance of the virtuoso, musical criticism and passionate listening.

Symbolism. The main function of the great musician is to make his music symbolic. He must take the listener out of the humdrum attitudes of life through the avenue of musical feeling into a state of abandon, which, in its extreme form, approaches ecstasy and obliviousness to material surroundings and facts. Even the devices of program music give but remote and stilted aid. His function is to enable the listener to live the art as he himself lives it symbolically. In this respect he differs from the sculptor and the painter who, while cultivating this symbolic attitude, are clearly held to the necessity of utilizing objective realities. It is not easy for the musician to take himself out of this mood. At the moment that he talks shop and business, the symbolizing habit is constantly pressing in upon him. Through his mastery of the symbolic life he feels rich, exclusive, powerful, and self-contained. Some people think that is queer.

Precocity. Since, as a rule, the musically gifted are proportionately precocious, they begin early in childhood to realize their peculiar gift for

musical appreciation, individual interpretation, and often fabulous child performance as more or less child prodigies. This tendency to become a prodigy is inherent in musical precocity and starting early makes him conspicuous as a child, interferes with his adaptation to the behavior of the common man, and leads to a specialization and intensification of those elements which gain for him the approbation of his constituency, a following for the hero, and towering admiration. On the other hand, it makes him an object of ridicule.

Usually it is the narrowly-educated musician that is strikingly temperamental. To excel in his art, supreme effort has often centered upon that goal at the sacrifice of a broadening education and the development of a well-rounded and healthy-minded personality. Current academic recognition of music and improved facilities for training will decrease the number of temperamental musicians. After all, temperament is not all a gift: it is largely acquired through learning in the school of hard knocks. It is often a cheap imitation.

What then is the musical temperament? It is a species of the artistic behavior found in all artistic pursuits, arising partly from heredity and partly from training, environment, and simulation; intensified by high sensitivity, highly-strung disposition, dominant ear-mindedness, emotional strain, lopsided education, pursuit of esthetic goals, leadership and hero worship, and often a forced precocity.

This type of analysis could be carried much farther, but these items may suffice as fair samples to show that the artistic temperament in music is an essential gift demanded by the nature of the art. It may be good or bad, inborn or cultivated, genuine or simulated, and often runs into eccentricities so that we frequently view it in a superior attitude of amusement. But let us thank all the gods in the Kingdom of the Muses for their great gift, the potentially good musical temperament.

THOUGHT REVIEW

General Principles

- (1) The musical temperament may be good or bad.
- (2) It is inherent in the intensive music situation.
- (3) The musical temperament generally has a hereditary basis in a highly-strung nervous disposition.
- (4) It has large common elements with the artistic temperament of the painter and the poet, but is more eruptive.
- (5) It is often of strategic advantage in musical performance.
- (6) In many respects it is a defense reaction.
- (7) It may be either extrovert or introvert.
- (8) It is most fascinating when artistic and serenely serious; most ludicrous when affected; and most abominable when it takes the form of tantrums and exhibitionism.

Consider These Questions

- (1) Can you name ten musicians who exhibit a "charming" musical temperament?
- (2) Can you name ten musicians who exhibit a "noxious" musical temperament?
- (3) Can you recall ten practical jokes which exemplify temperamental behavior, good or bad?
- (4) What are some of the first evidences of temperamental behavior in the musically precocious child?
- (5) Should the musical temperament in a child be cultivated?

- (6) Why is the current music supervisor rarely temperamental?
- (7) Was long hair a temperamental trait?
- (8) Is temperamental behavior proportionate to excellence in music?

Discuss This Situation

When the director of the professional symphony orchestra faces a group of temperamentally hardened performers in rehearsal it is war to the finish—victory or defeat. Recall some characteristic historical instances of artistic strategy in such a situation.

Chapter VI

MUSICAL INHERITANCE

The whole problem of mental inheritance is in the air, both in the sense that it is current and in the sense that it is relatively intangible. The struggle is best illustrated in the current approaches to the problem of inheritance of intelligence. In this the geneticist has not got far from base, but much has been learned in regard to the nature of the issues involved. In the field of music the geneticist has approached the subject experimentally without understanding the musical life; and the musician has approached the matter practically without being a competent experimenter. The psychologist has certainly not done his duty in clarifying the issues. The most pressing need at the present time is for such clarification. This can not be the work of one man or one generation, but must be achieved through co-operation of both sides in order to clear the way for valid experimentation.

In order to indicate the character of the problem we are now facing, I shall first venture to state some fundamental assumptions upon which probably all competent investigators agree and, second, venture a little way in the direction of identifying concepts of musical life which can be dealt with experimentally.

ESSENTIAL PREMISES

The mechanism of heredity lies in a single germ cell carrying the character-determining chromosomes which consist of organized chains of genes. In the character and organization of these genes in the fertilized cell we find the complete "blueprint" for the future individual in so far as it is to be determined by heredity. In the twenty-four pairs of chromosomes in the fertilized human germ cell we find the long and diversified heritage of each parent represented through the union of the sperm and the ovum. The selection and the organization of the genes in these chromosomes adequately represent what the future individual can be.

This genetic constitution is modified by the cytoplasm, the supporting part of the cell which is its first environment, and further by the entire embryonic environment. Any changes that take place after the launching of this cell, whether before or after birth, are regarded as environmental. In the embryonic life, this germinating cell develops by processes of cell division and specialization into the complete human organism ready to function more or less immediately after birth. This heritage has fabulous resources in the form of possible facilities for future development. As nature was prolific in the storing and transmission of countless hereditary characters in the genetic constitution, so the equipment of the child at birth is astonishingly prolific in the provision it makes for diversified development of the individual. Development from this stage on must, therefore, of necessity take place through a process of selection and specialization in which certain characters are given right of way and many are subordinated or inhibited by conflicting interests, but the great mass remain relatively latent or dormant. We may assume that superior musical talent is determined in large part by superior musical heredity, and that inferior musical talent or lack of talent may be determined in large part by a correspondingly defective heredity.

The science of heredity in the strictest sense focuses upon the study of the identification and organization of the genes in relation to the determination of characters which shall appear in the genetic constitution and determine future structures and functions of the individual. When the geneticist deals with specific anatomical structures, this relationship is traceable with comparative ease; but when he comes to deal with more or

less complicated physiological or mental functions, the tracing of this relationship becomes rather baffling on account of the complexity of the final product.

Turning then to the issues involved in the interpretation of musical inheritance, we must face certain theoretical assumptions. One of them is that a scientific study of musical heredity cannot be pursued on the assumption that mind and body are two distinct entities, each inherited independently. Nor can we hold the old doctrine of psychophysical parallelism. All human genetics proceeds on the assumption that the human individual is one psychophysical organism. Our musical experience, observation and measurement will therefore represent views from the mental side; our organic studies may be views of the same things from the physical side.

Furthermore, musicality is not one specific human trait but an infinite hierarchy of traits running through the entire gamut of the psychophysical musical organism. To make any progress whatever, the scientist must make the supreme sacrifice of attempting to deal only with specific isolable factors apparently small and remote in themselves. The situation is analogous to that of purely physical features. It is generally admitted that the structure of the physical organism is heritable. But when we show that the color of the eyes of the fruit fly is heritable and that this inheritance takes place in a very complicated way, as has been adequately shown, we have simply identified parts of the structure and function of the genes in one specific feature in the vastly complex physical organism, however fundamental and characteristic this particular feature may be. This analogy applies in principle to the genetic study of the musical life. The crux of the difficulty lies in the identification of heritable factors.

Again we must remember that the musical mind is first of all a normal mind, a normal psychophysical organism ready to begin to function immediately after birth. What we shall look for then in a psychophysical organism is the presence of certain resources especially favorable or especially unfavorable to the normal functioning of the musical mind. We may assume that an average capacity present in the genetic constitution may be adequate for musical purposes but that exceptionally gifted persons require these traits in a correspondingly exceptional degree and that exceptionally unmusical individuals lack essential elements. The most wonderful thing is that a person can come into the world with a musical

constitution at all, but the problem of heredity centers around individual differences, and these are more easily approachable than the total function. As in genetic studies of the inheritance of color blindness it has been possible to identify types, so in musical hearing we may look forward to the identification of types of defect and types of superiority deviating markedly from the normal.

Common observation and reasoning convince us without question that musicality is inherited in some mysterious way and this follows also from general considerations of current theories of biological inheritance. But when it comes to the scientific determination of laws of such inheritance, we face high barriers. Biological laws of inheritance must be established in terms of the genes; a specific biological structure or function must be related to gene organization. Let us call this measurement of the first order. Such measurements are most readily applied to anatomical structure and physiological function in the neuro-muscular organism. This is notably clear in the anatomy and physiology of the ear and its connections. It is equally applicable to the anatomy and physiology of the vocal organs—the bellows, the vibrators and the resonators for voice. It is conceivable, for example, that the length, the mass, the mode of attachment, and the general position and shape of the vocal cords and the mounting of the voice box are heritable characters traceable to genes and referable to musicality as the physical organs for voice.

We can also find relationships to the endocrines, which are in large part the determinants of musical emotionality. Electro-physiology is now giving great promise for the identification of functions in the ear and the brain and its central connections and is establishing interrelationships. Many of the laws of heredity established by measurements of this order probably refer to fundamental biological principles of inheritance in the psychophysical organism as a whole. By a physiological analysis of the sensory, motor, and central factors which operate most significantly in music, the systematist can set up a respectable body of biological facts in regard to musical inheritance which are antecedently probable in terms of the functions of genes and result in the structure and function of the musical organism.

Since the medium of music is sound, we shall look first for an exceptionally responsive or unresponsive ear, including not only the physical ear but the central organs in the nervous system through which it functions. This is basic for two reasons: First, because it determines what

stimulation from the world of sound shall enter into the experience of the musical individual to a high degree; and second, because the purely physiological receptivity or organic response to sound acts upon and modifies the state of well-being or ill-being according as the auditory impression is beneficent or noxious in so far as its acts upon our circulation, metabolism, temperature and other organic processes. Such well-being or ill-being is, of course, in part the foundation for the feeling of musical pleasures and pains.

If we would gain a true and comprehensive insight into the nature and extent of role of environment in musical life, we must start with some established facts or reasonable assumptions of what is "given" for environment to act upon. The heritage is the capital fund which the environment invests or squanders. Only by knowing the hereditary contributions can we appraise the environmental contributions. In the study of the fruit fly, for example, the revelations of factors which must be regarded as environmental are quite as significant and essential as the revelations about the original organization of genes. The determination of the limits of heredity is the best means for revealing the functions and possibilities of environment. The music geneticist will therefore learn fully as much about environmental influences as he will about hereditary influences in studying heredity.

PSYCHOPHYSICAL MEASUREMENTS

The music geneticist can approach many significant aspects of the subject through psychophysical experiments for which we now have fairly standardized procedures. For the present purpose, we may call this measurement of the second order as compared with the anatomical and physiological measurements. It proceeds out of, and is a complement to, the anatomical and physiological foundations and probably represents the most fundamental approach from the psychological and musical points of view. These measurements deal primarily with sensitivity and discrimination on the sensory side and the corresponding processes on the motor side. Among them we may recognize two levels: The simple or elemental, in which a specific mental process is related to a relatively specific organic basis; and the complex, which relates to co-operative functions of the elemental capacities. Of the former we have four; namely, the sense of pitch, the sense of loudness, the sense of time, and the sense of timbre—each of which is correlated with a specific attribute of the sound wave, which is the musical medium. We have basic measurements of the hearing of rhythm, consonance, volume, and sonance—all of which represent relatively complex patterns. Each of these complex functions has a unitary character. Rhythm, for example, is not merely time plus intensity; it possesses a unitary character. Because of the difficulty of dealing with the complex patterns, precedence should be given to the four elemental or basic capacities. Excellence in these capacities contributes toward ear-mindedness, of which the auditory image is the most specific characteristic; but at the present time we have no adequate objective method for the measuring of auditory imagery.

On the motor side we have corresponding measurements of speed and accuracy in the motor control of each of these factors represented in the sound wave; namely, frequency, amplitude, duration, and form.

The term elemental should be used with caution because we never encounter a purely elemental state or process. Even in the very simplest form they are merely more or less specific phases of the mental organism; and at any level at which they are observable they probably involve environmental accretions. It is the old story: We never experience pure

sensation but meaningful perception. Yet under the most careful experimental control the identification of such specific functions may be reasonably reliable and have considerable validity.

Adequate measurements of the sense of timbre are new and therefore have not been employed extensively up to date. But the sense of pitch, the sense of time, and the sense of loudness, together with the sense of rhythm and immediate tonal memory, have been used extensively.

The significance of such measurements depends upon the rigidity of the scientific technique and the selection of subjects for experiment. Reliable measurements have been made on a variety of groups and for different purposes more or less related to the problem of inheritance. Studies have been made upon musically precocious children to determine to what degree they were gifted in each of these capacities. All the available blood relatives of six of the foremost musical families in America and a number of such families in European countries have been investigated. These capacities have been measured in selected virtuosi in various fields of music. The measurements have been used for the determination of qualifications for musical organizations and for the analysis of admissions to music schools. Simplified forms of the measurements have been made upon very young children in musical families. Numerous cases of failure in musical education have been investigated and often explained on the basis of presence or absence of these basic capacities. Surveys have been made on groups representing highly-privileged or under-privileged children in the matter of musical facilities. Some of these measures are now a part of the standard tests and measures administered in the public schools so that comparisons can be made with blood relatives, and data are becoming cumulative for scientific comparison of successive generations. Numerous racial studies have been made on a large scale, comparing these capacities, for example, in different degrees of race mixture—as in the transition from pure blacks through mulattoes to whites in a large Negro community, or the comparison of racial groups in Hawaii, the school children in different European countries, Indians with whites, and distinctive races and primitive peoples in different parts of the world.

From this large array of facts certain findings seem to be significant, taking these measurements as a group. First, the sense of pitch, the sense of loudness, and the sense of time reveal no distinctly significant differences in racial groups, in culture-levels, or at age-levels, when adequately measured.

In many cases this holds also for the sense of rhythm and tonal memory. This is probably indicative of the fact that the basic capacities for hearing in individuals now living and capable of being tested adequately are physiologically at the same level. This conclusion is in harmony with the observation that these capacities which function in music, function also in the vast varieties of orientation through sound at all levels of man now living. It is also analogous to what has been found in vision. Second, it develops that in each and every one of the groups studied there are enormous individual differences in each of these capacities and that the extent and distribution of these differences do not differ significantly from what we find in the public school children of the United States. Third, where comparisons of capacity and achievement have been made reliably, it has been found that those who have achieved distinction in music have these capacities in a significantly corresponding degree; but much larger numbers of those possessing superior capacity who have not been discovered as musical, either by themselves or in their environment, are revealed. This fact rules out many of the statistical studies of heredity in terms of musical achievement. Fourth, these capacities represent relatively independent factors in hearing. Fifth, marked superiority or inferiority in these capacities is of predictive value for musical achievement and guidance in education.

On the motor side but little progress has been made. Principally because the measurements are laborious, significant elements have not been identified, and moderate motor capacities in speed and action are adequate for most musical achievements. Daily observations reveal that children may be slow and accurate, slow and erratic, fast and accurate, or fast and erratic in various degrees and combinations. It would, however, be of musical significance to discover to what extent and in what manner these traits are inherited from generation to generation.

In view of these discoveries, it is evident that there is some material available for technically rigid genetic interpretation in terms of currently recognized principles of inheritance. All the records on the six foremost musical families of America are available in the confidential files of the Carnegie Institution, at Cold Spring Harbor. Highly reliable measurements on all the students in the Eastman School of Music for the last fifteen years are available. Various public schools have vast cumulative data, and elaborate collections are being worked upon in the Winderen Laboratory, at

Oslo. But with the exception of the Carnegie Institution and the Oslo collections, adequate measurements of whole families are absent.

What is needed now is a thoroughly reliable series of measurements on entire musical families and the interpretation of these by a thoroughly competent geneticist in terms of established biological principles of inheritance. It is especially important that both parties shall be competent to take into account the numerous lessons which we have learned from the extensive efforts that have been made in the attempt to measure the inheritance of any mental trait, such as human intelligence. In the human situation we cannot breed successive generations rapidly, as in flies or mice, for experimental purposes. We must, therefore, economize time and effort by taking the most readily available material. For this purpose I have suggested three possible methods (*Psychology of Music*, McGraw-Hill, 1938). The first is that we start with the highest 10 per cent and the lowest 10 per cent in an adequate sampling of fifth-grade children in a school system and work back by making the same measurements on the available blood relatives of these two groups. In effective organization much time can be saved by making group measurements in a co-operating community, such as a city ward. A second procedure would be to secure an adequate sampling of musicians and measure forward and backward to cover three generations in which the matings of musical and unmusical parents could be traced. A third procedure would be a systematic collection of measurements on school children for a generation or more giving special attention to the showing of blood relatives. We cannot, however, stress too strongly the importance of having these measurements made throughout by an experimenter thoroughly competent in this field and the equally thorough biological treatment of data by scientists thoroughly competent in that specific field. If a biologist wants to start the ball rolling from his point of view, the records in the Eugenics Record office of the Carnegie Institution furnish a fair and reliable sampling.

In proposing this conservative approach through psychophysical measurements, I do not wish to belittle the insight, common knowledge and theories of inheritance which have been obtained by observation and statistics in terms of musicality as a whole—as in biography, autobiography and letters of great musicians or in the study of musical families. But we are confronted with the fact that these deal largely with unanalyzed situations so completely covered by factors of environment and training as to make

them useless for strictly scientific purposes. Nor would I belittle the significance of general traits, such as musical intelligence, creative imagination and the artistic temperament, or facilities for specific skills, such as sight reading, and the memorizing of repertoires. We know a great deal about these and unquestionably have the right to assume that they have an hereditary basis. But scientific studies in heredity may be more properly approached through the simpler and more elementary capacities.

For scientific purposes, we cannot, of course, mix basic measurements and current ratings of musical achievement. There have been numerous approaches to this subject from the musical-achievement point of view, and these have furnished many suggestive leads and probably point to unquestioned facts about the inheritance of musical talent. But the science of genetics rightly rests upon and demands the isolation of specific factors which can be measured; and for that purpose the musical geneticist must, for the present, sacrifice many otherwise interesting approaches from the point of view of rated achievement and be willing to await the laying of foundations of rigidly conducted measurements which can be described, interpreted, and verified.

THOUGHT REVIEW

General Principles

- (1) The question is not whether or not musical ability *is* inherited, but *how* it is inherited.
- (2) The musical heritage through each parent is transmitted through a single germ cell.
- (3) The mechanism of musical heredity lies in the organization of the genes in the twenty-four pairs of chromosomes found in this fertilized germ cell.
- (4) Scientific principles of musical inheritance can be established only through experiment and measurement of specific capacities; they cannot be derived from studies of musical achievement alone.
- (5) We come into the world a fully-equipped organism which is evolved from the germ cell before birth.
- (6) This hereditary equipment is prolific in possibilities; at best, the musical life can develop only through a small part of these inherited resources.
- (7) Defects in the inherited equipment set corresponding limits to musical achievement.
- (8) The scientific approach to the problem of musical inheritance can be made through psychophysical measurements of musical capacities with biological interpretation in terms of genes.
- (9) There is a considerable body of general biological principles of heredity which probably apply to the inheritance of musical talent.
- (10) Scientific study of the nature and the role of musical environment must begin with knowledge of what is "given", the original hereditary stock.

Consider These Questions

- (1) What bad effects in education come from overemphasis of (a) heredity, (b) environment?
- (2) What educational advantages will accrue to the child from a sound and balanced recognition of the role of (a) heredity, (b) environment?
- (3) What would be the advantage of knowing scientific laws of musical inheritance?
- (4) What is new to you in the author's proposal?
- (5) Do you accept the author's theory of the relation between body and mind?
- (6) Why can we not derive laws of musical inheritance from statistical studies of musical achievement?

Discuss These Situations

- (1) A music teacher says "Give me a normal child and I will guarantee to make him musical." What qualifications would you suggest?
- (2) A teacher has measured a child's sense of pitch and found it superior. Should she say "You are highly musical," or "You have a superior sense of pitch?" What is the difference?
- (3) An investigator has recently collected interesting material by interviewing (a) thirty-six of the outstanding musicians of the world, (b) the entire cast (thirty-six) of the Metropolitan Opera Company, and (c) fifty selected graduate students in the Juilliard School of Music, about their parents, and other blood relatives and has attempted to interpret the findings in terms of genes. Granting that his theory of genes is correct, what is it that limits or invalidates the procedure?
- (4) Long ago the author investigated the musical ability of three children in a minister's family and found that the oldest daughter rated very low on basic capacities and that she had not made satisfactory progress in eight years of musical training. The younger sister rated

very high in musical capacities and had made splendid progress with but a few lessons. The brother, the youngest of the three, was like the younger sister. The minister held that where the great Creator had made the children unequal, it was his business to make them equal, and so he sent the eldest daughter to the New England Conservatory of Music where she was accepted and graduated and now holds a diploma, though, in her community, a musical nonentity.

- (5) Richard Bach Smith is a musical prodigy according to his mother. At the age of five he plays, sings, composes, and is an accomplished sight-reader. Is this heredity, environment, or both? How would you order his musical education?
- (6) Miss Jennie Lind Jones has a musical father and an unmusical mother, lives in a fine musical environment, and has enjoyed fourteen years of excellent training; but she "has no rhythm," her piano phrasing is "cold", and her music is a "drudgery." What do you predict for her? What would you have done for her?
- (7) Subject for debate: Given both, resolved, that inherited talent is a larger factor than musical education and environment in the development of a great musician.

Chapter VII

THE FUTURE OF MUSICAL INSTRUMENTS

Are we nearing the end of the "horse-and-buggy" stage of musical instruments? Can the possibilities for revolutionary procedures now looming up in the construction of musical instruments be as strategic for music as were the principles embodied in the coming of the automobile and the airplane for transportation? Those of us who remember that faithful animal and servant of man, the horse, and the conveyances he served, look back with fond appreciation upon what amounted to a sort of fellowship with a fine-performing animal and the luxury of being conveyed by him in saddle or on wheel. So future generations may look back upon the past in fond memories of the companionship they have enjoyed with their favorite instruments, which may be destined to a niche in the historical museum. But in spite of competition, the horse has survived, and so probably will the fiddle and some of its companion instruments.

It is now safe to predict that the future instrument maker will be able to produce any sound now known in nature or in art that may possibly have musical significance. We already have at hand the means by which any such sound can be adequately defined, described, specified, measured, analyzed, and reconstructed. And there is reason to think that with the conquest of new and marvelous resources for musical media, musical composition will move with strides in step with instrument building.

The musical devotee is, therefore, facing new issues, thrilling and possibly heart-rending. Can a musician adapt himself to these changes? Will he tolerate modifications of old instruments, radically new creations of instruments, revolutionary new types of ensembles, and radically new types of musical creation? Can musicians adapt themselves to these new musical media and musical forms as rapidly and completely as we have adapted ourselves to the transition from horse and buggy to automobile and airplane within the span of less than half a century? The answer is probably "no", for

good reasons. Yet, sooner or later, the transition will come in the form of new musical media, new musical composition, and new types of musical appreciation and attachment.

POSSIBLE LINES OF DEVELOPMENT

We can now foresee that musical instruments will be submitted to critical analysis, with improvements even on the very best; that substitute forms in great variety may be developed for any now available musical instrument; that new instruments will be designed for the production of new tone qualities and other musical effects; that new ensembles may be built for any number or kind of instrument, so that it is within the bounds of possibility that the entire performance of the symphony orchestra, the symphonic band, and the grand opera may be performed through a single instrument operated by less than half a dozen persons; that the transmission of music by remote control of the instrument has extraordinary possibilities; that a vastly superior control of tone for precision and modulation can be realized; that the cost of musical instruments may be greatly reduced; that the number of players needed in ensemble performance may be reduced, since, on the analogy of the pipe-organ player, one individual may perform for an entire orchestra; that current music which has been hampered by limitations of the instrument may be perfected and new types of music may be introduced; that the musical instrument may become a medium for the production as well as the reproduction of song and speech; that the musical tone may be associated with other esthetic appeals, such as visual presentation of color, relief, and dramatic action—these are all within the realm of possible predictability.

The range of possibilities may be illustrated in the case of a generator for a single tone. A few years ago a graduate student came to me and said that he wanted to take as the subject for his doctoral dissertation the building of an electrical organ. I told him that if he would build a generator for a single key I would assure him a stipend for three years and all the needed facilities of the Iowa acoustical laboratories. He accepted and made good to an extraordinary degree. The tone generator he built is composed of the first sixteen partials, *i. e.*, a fundamental and fifteen overtones in harmonic series, each virtually a single pure tone. The number of partials, the form of distribution of partials, the amount of energy contributed by each partial, the phase relationships of the partials, and the fundamental pitch are under control and may be set up in any combination. With this number of

variables, mathematicians will say that any desired tone quality of harmonic structure up to more than a million kinds of tone can be produced. The ear cannot hear all these differences, but the instrument provides keys for as many steps in the entire series as may be musically significant, and the tone at each of these steps can be specified for production, described, and repeated indefinitely. Thus, it is possible to make the instrument speak any vowel in so far as it involves harmonic structure, and the harmonic composition of any musical instrument may be imitated. Provision for inharmonic elements and noise accessories which are necessary both for vowels and instruments can, of course, be installed with this generator. Given one tone with such a range of possibilities, it is but a series of logical and fairly simple steps to provide a complete musical instrument by simply multiplying notes of this kind. This is the type of development we see now in a variety of electronic instruments.

The improvement of existing instruments. Musical instruments now in use can be improved. We now have the means for the technical analysis of the character of the tone produced by any instrument as a whole or by any particular feature in its construction, so that faults and limitations can be definitely allocated. Recent investigations have revealed faults in the best of violins. Some of these faults or limitations may be corrected by change in construction. For practical purposes, a 1939 violin of American make may approximate the good qualities of the Stradivarius, and there is no doubt that improvements could be made upon the famous old instrument. The same is true in principle of all individual musical instruments now in use. One of the obstacles, however, will be the unwillingness of many musicians to face the innovation of change in the looks of their beautiful instruments so tenderly loved and guarded.

The piano, as we now have it, has limitations and defects, some of which can be reduced or eliminated by the adoption of new principles of action and activation, and by construction according to acoustical specifications, based upon the measurement of the effect of each feature upon qualitative and dynamic values of tones. For example, the characteristic tone of the piano as distinguished from most other instruments comes to a dynamic peak immediately after the hammer stroke and falls off rapidly. We have learned to make allowance for this so that in hearing music we have a tendency to hear a quarter note as of a given even loudness although the sound fades off sharply during that period. It is well known that with the

drop in the loudness of the tone there is a corresponding change in the quality of the tone, which change again we have, of necessity, practically learned to ignore. If it should prove desirable, it is now quite easy to provide a mechanism which will sustain the piano tone at even loudness, and therefore uniform tone quality, during the time value assigned to it. The distribution of resonances of the instrument can be greatly improved by balancing. The necessary mechanical noises which accompany the production of tone in a piano may be largely eliminated if that should prove desirable. The acoustical engineer can now point out dozens of features in the piano which might be improved in future construction based upon analysis of the output in sound. In the same manner the organ, the king of instruments, if it is to maintain its pre-eminence in competition with substitutes, demands improvements in the light of new facilities. The more complicated the instrument, the more possibilities there are for improvement. During the last hundred years there has been a steady betterment in the mechanisms of practically all the leading musical instruments; but this movement will rise to great heights in view of the new tools for investigation and new materials and principles of construction.

New substitutes for existing instruments. Substitutes for all instruments now extant may be expected to come mainly through the development of electrical construction, although many forms of mechanical devices may be used independently or with the electrical. There is no question at all but that with such resources, substitutes for a stringed instrument, a wood wind, a horn, and many varieties of traps and accessories can be built so as to embody increased musical resources. The principles for the construction of such a violin, flute, trumpet, or any other single instrument are already in hand. The construction may embody such mechanical devices as strings, membranes, and pipes; but if so, these will be electrically energized. We may predict that substitutes for single instruments will increase in great variety and that very simple forms adapted to age and advancement in the playing of specific instruments can be supplied. The variety of means for stunt performances may be increased to an alarming degree.

New ensembles. Another significant line of development will undoubtedly appear in the matter of ensembles. By the utilization of electrical construction, a single series of generators may supply the harmonic structure of a tone from each and every instrument now in use. Thus, a bank of sixteen violins may be supplied from this single source and

sixteen individual violin tones may be played in identical pitch, even dynamics, equal temporal movements, and uniform tone quality. Such uniformity would however be of limited musical significance. The important thing musically is the fact that in such a bank any desired form of artistic deviation or differential enrichment may be provided for. The same would apply to wood winds, horns, bells, drums and other percussion instruments in large part. Where further representation of mixed tone, inharmonics, or noises are demanded, they may be added. The substitute for a drum can increase the precision and range of the drum sound without the use of the drum, as, for example, in the present "Novachord". Thus, it is conceivable that with the exception of certain unforeseen limitations, the instrumentation of an entire orchestra or band can be built into a single unit operated from a single console.

From what we have seen in the way of marvelous demonstrations in recent years it is reasonable to suppose that entirely new types of complex instruments will be invented, bringing to music hitherto unknown resources. Furthermore, with the instrumental music as such, provision may be made in the instruments for words in speech and song, and the visual presentation of dramatic action in color, relief, and movement which may be controlled from the console. The goal of embodying in a single instrument or coupling units of instruments the means of performing chamber music, orchestra, band, and the grand opera, is without doubt no wilder prediction than was the prediction of radio or airplane a few years ago.

NEW MUSIC

The improvement of old instruments and the introduction of new ones will call for an unprecedented revision of old music and a creation of new. When music was written for the well-tempered clavichord it was limited to the resources of that instrument. The same is true of music for all instruments. The music had to be limited to the available resources of the instrument. It is reasonable to suppose that composers will respond from time to time with up-to-date adaptations and new creations, taking advantage of each of the new resources for range of pitch and loudness and new resources for variety in harmony and richness of tone. It is equally conceivable that the composer may set up new demands to which the inventor and instrument maker may respond on call. It is difficult to realize what extraordinary enrichment in musical resources may spring up under the impetus of new instrumental resources. There will be new treatments of scales and intervals, since the pitch control will be far more flexible than it has been. Perhaps one of the largest innovations will be in the freer use of intonation not built on any particular scale but soaring with the greatest freedom on an instrument as we now hear it, for example, in the singing of Negro spirituals. Performance scores show that these natural singers defy scales, but produce beautiful effects through their free and soaring pitch inflection. Stringed instruments have been hampered by accompaniment and by tradition and theory. We can anticipate significant developments outside of our diatonic scale which has come to be a sort of strait-jacket, at least theoretically. It has been shown, for example, that a quarter-tone instrument is not of much use unless music is written not only for these intervals but in modes, themes, and atmosphere adapted to such purpose. The pitch range of the composition will be extended; so also will the dynamic range. Countless new features can be introduced for enrichment of tone and variety of harmony. Nomenclature will develop so that the composer may not only think in definable terms but may be able to inject new elements of terminology into the score. For various types of ensembles the music will, of course, have to be written or adapted specifically. Stunt music will here find unlimited opportunities for novelty and escape from conventional tone. This may give us relief from the limitations of jazz and

swing, which have been so boring in recent years. There will undoubtedly be great bewilderment as to the limits of tolerance for new media and new forms for musical creations. History has revealed clearly that the adaptation of taste and tolerance requires time, and conservatism is often a beneficent safeguard.

The most fundamental recent achievement in the building of new instruments is that of producing pure tones as well as rich tones in perfect harmonic structure without accessories. But for various reasons music has always utilized more or less inharmonic structures and has imitated, or at least tolerated, noises in the forms of hisses, twangs, and all kinds of inharmonic and noisy distortion. Recourse to discord will always be demanded in music; but it is a question as to what extent aperiodic sounds or noises are necessary. We have learned to accept the hisses, scratches, rattles, thuds, and bangs of countless varieties in musical tone. Some such accretions are present in the tone production of every instrument now in use. They have arisen as impediments in construction, but have come to be accepted as characteristics of the instrument and therefore have added greatly to the individuality of each instrument. Will the future musical public insist on having these or will there be a feeling of relief when we can get rid of them? The answer is probably in the middle ground because music, at least as we know it now, calls for a great variety of noises in the interest of realism. Possibly in the future, noise may be given a chastened and more honorable place in the family of tone qualities.

The significant thing to note here is that in new types of instruments any kind of crude sound or noise can be introduced at will. This will answer the purpose of program music in that perfectly realistic reproductions can be made of the sound of steps in marching, the slam of a door, the squeal of a pig, the bleat of a lamb, the crow of a rooster, the roar of an airplane, the sigh of the wind. A comparatively small outfit of sources for accessory noises can serve a great variety of purposes in the hands of the artist. New responsibilities for the composer in these respects may be foreseen.

PLAYING

As in the automobile, for example, the tendency is to make everything automatic and relieve the human control of effort, we are moving fast in the direction of creating automatic devices in musical instruments. It is safe to predict that a variety of instruments for children will be more easily played than at present, and that even for the virtuoso a number of the factors which have been difficult to control will be simplified and mechanized. This will be particularly true in ensemble instruments where a single player may control a large number of instruments or where three or four players, of whom one is the conductor, may render the equivalent of an orchestral performance. Barring the limitations of the instrument it is significant that in such performance the conductor will have vastly superior control of the situation in that he is in direct manual control of all those factors which in the present orchestra he tries to control through the medium of the individuals and masses of players with the baton as a sort of whip. In the new orchestra, he will sit at a panel with levers and buttons through which he will be able to control the interpretation he desires to make.

This conducting may be done with meticulous precision and apparently magical result through remote control, as was illustrated, even with the use of present instruments, when Stokowski sat at his panel in Washington, D. C. and controlled the performance of his orchestra in Philadelphia.

SPECIFICATIONS FOR INSTRUMENT CONSTRUCTION

The musical medium whether in art or nature has four, and only four, elements; namely, pitch, loudness, time, and timbre. These are four characteristics which may be expressed in terms of the sound wave, as frequency of waves, intensity of waves, duration of waves, and form of waves. The recognition of this fact vastly simplifies the mastery of tone production in musical instruments. Let us consider the factors which may be classified under these four heads.

Pitch. New devices for the control of pitch make possible very superior control of the pitch in pitch precision, flexibility, linearity, range, varieties of scale, and harmony. Instruments can be tuned to a higher degree of precision, retuned more quickly, and kept in tune longer by the new means of tone production. Therefore, intervals can be made more exact, the pitch can be sustained with precision, any pitch aspect can be described, desired effects can be specified, and artistic adjustments can be made for musical effect.

Any desired type of flexibility may be introduced, such as the gliding attack and release, periodic or progressive changes in a sustained tone, as in the vibrato, or rise and fall in tone required during the sounding of a single note. The glide, the portamento, and a variety of other pitch ornaments may be produced. Adjustments from any size of intervals for artistic effect, such as a quarter tone or enlarged or diminished intervals, may be produced. In short, the pitch inflection can be controlled either within a single note or a sequence of notes, and just intonation or tempered scales of various sorts may be produced. The possibility of combining individual notes in harmony is unlimited. The pitch range of the instruments can be carried to any height above the present gamut of instruments, even into the region of supersonics, which may possibly have physical effects on the feeling of well-being or ill-being of the musical organism.

Loudness. The range of loudness can be extended both in the direction of increased loudness and perfectly controlled softness. The loudness of tones throughout the entire range can be equalized. All forms of dynamic shading and fading and differential steps of loudness can be introduced for artistic effect. The dynamic linearity, as in the organ, can be sustained and the

fading effect, which distinguishes the piano from the organ, can be adjusted to any degree. All present pedals as well as new pedals may be introduced.

Time. Through the precision made possible for various forms of automatic action in the instrument, all temporal aspects of tone can be enhanced by precision, controlled tapering at the beginning and end of the tone, various forms of synchronization, and the balancing of the roles of time and stress in rhythm may be achieved.

Timbre. The most significant new feature is, of course, the extension and control of richness in timbre and sonance, which constitute tone quality. The range of timbre changes may extend from a pure tone to the inclusion of even more partials than the ear can hear, possibly thirty or forty, with an infinite variety of harmonic forms of distribution between these extremes. Great innovations through inharmonic and aperiodic sounds are coming. It is now possible to maintain a uniform timbre which gives an element of dignity and stability to the tone in a sustained note. But since deviation from the regular is of far greater importance, flexibility of tone quality is even more desirable. This flexibility in timbre for the duration of the tone which determines tone quality, technically called sonance, may be provided for by various devices. For example, in the present "Novachord", the various forms and degrees of vibrato are at the command of the player and are based on scientific investigation leading to specification for this musical ornament. Given the mechanisms for harmonic structures and inharmonics, we need only the addition of rough noise of various kinds to make the tone realistic. The timbre available in the musical instrument will therefore run from the pure tone to the roughest noise.

In the above I have attempted to present a picture in rough outline indicating present trends and predictable futures for music through the improvement of musical instruments. The instrumentalist can point out many features not here mentioned. The musician may yearn for new resources in his musical medium for which he can now lay down specifications. The listeners must be prepared for startling innovations and thrilling new sources of pleasure in music. It will take generations to finish the theoretical picture I have here sketched in bold outline—but let us look forward to a progressive unveiling and revelation of possibilities. But take no alarm—the horse, if not the buggy, will survive.

THOUGHT REVIEW

General Principles

- (1) We are entering upon a new era for musical instruments.
- (2) Present instruments will be improved, new instruments will be innovated, new ensembles will be achieved.
- (3) Conforming to this progress, present music will be adapted and new types of composition will appear.
- (4) In this changing order musical tastes will undergo significant changes.
- (5) The composer may now specify new conceptions of desired features, and the technicians will provide them.
- (6) Instrument makers will be able to produce or reproduce any musical sound in nature or art.

Consider These Questions

- (1) If the violin strings are mounted on a perfect resonator with different shape from the present violin, will that be a violin?
- (2) If the piano can be built so as to eliminate all accessory noises, will pianists like it?
- (3) If the playing of new instruments will be made easier and better, will that tend to increase the number of students in music?
- (4) If all the good characteristics of present musical instruments can be reproduced in a much smaller number of instruments, will that be welcomed by the musical public?
- (5) If we are to have quarter-tone music, which is now possible, why will this demand new music aside from the size of the interval?

- (6) If an electronic organ could be built to do all that the best pipe organ now does, if the visible part were reduced to the appearance of a little writing desk, and if speakers were concealed in the walls throughout the building, how would the worshipper in church react to that situation?
- (7) Children now build their radios from purchased or home-made parts. It will be still easier for them to build a variety of musical instruments. Will they do it and with what effect?

Discuss These Situations

- (1) At the Riverside Laboratory, Colonel Fabian built an organ with the pipes distributed throughout the three stories of a large building. The effect was as if the whole building were one grand organ. No matter how many sources there were for the same note, the tone of that note would always be heard as coming from a single source, the location of which depended upon the relative distances and intensities of the various sources. The result was "magical". Discuss the possibility and the significance of using this principle of installation in a house or in a cathedral.
- (2) It is possible to build a comparatively simple instrument which can yield pure tones throughout the musical register. Consider what role such an instrument might carry in an orchestra.

CHAPTER VIII

PRAISE AND BLAME IN MUSIC

Music is unique among esthetic appeals in that it demands immediate response in the form of praise or blame. The orator demands immediate response; but his appeal is to reasoning, not to feeling. The painter, sculptor, and poet demand esthetic response, but this response is delayed and does not keep the artist on the tip of his nerves to receive it. The musical appeal is all the more emotional because it is not only an appeal for personal recognition but for the aggrandizement of the noble art. The musician at all ages and all stages of advancement can perform for his own pleasure in isolation, but even here the demand for praise or blame on the spur of the moment is emphatic and essential.

It is, of course, fully recognized that at his best the musical performer is not conscious of making a direct appeal for approval by others. His objective is rather that of self-expression and experiment in the execution of an artistic skill. He is often aware of a negative attitude in his audience, and he may take an attitude of aloofness or superiority toward the audience, recognizing its incompetence. Yet back of it all, there may be even a distant hope of approval in a coming generation or the critical judgment of the select few; but the attitude of the immediate audience or the public toward his performance plays heavily upon his unanalyzed feelings and moods.

The bases for musical praise and blame are notoriously inadequate, be they meaningless approval, empty flattery, or censorious criticism. A general attitude of spontaneous approval without knowing or giving reasons among listeners in general is praiseworthy as an expression of good will; but for serious purposes we must question the competence of the listener for the response manifested. The fault may lie in the listener's ignorance of the art of music, his incompetence in recognizing elements of merit, his failure to credit the performer in relation to natural ability or purpose. Praise or blame may also carry or miscarry in so far as it may wisely serve to

motivate or discourage the performer on the specific issues involved in the performance. With the sudden popularizing of music, the incoming tendency is to associate music with the beginning of a scientific approach to the understanding of the music and the musician. This situation is changing at a gratifying rate for the good of music. It is worth while to consider in some detail the factors in this progress.

VANTAGE GROUNDS

Artistic insight. The extraordinarily rapid introduction of real music education in the grades and in the high school and college curricula, and even in graduate study, is changing the attitude of the listener by increasing his competence. Children and youth now hear more good music and hear it presented in such a way as to increase their understanding of it. At the college level music is presented as an academic subject, with primary emphasis upon the art of hearing music together with some knowledge of history and content of music. The training of high-school teachers of music has changed radically in a generation from the mere development of proficiency with an instrument or voice to a deeper insight into the nature of the art, its history, and its roles. At many levels well-developed courses in music appreciation have acquired a permanent status and have proved a good foundation for further penetration into the art of the appreciation of music. This implanting of points of view and development of esthetic attitudes in the academic instruction in the schools is hastened to fruition by the popularizing of music for the masses through radio and phonograph as well as through the penetrating of the skills acquired in school into the home and social institutions. The popularizing of good music, where public entertainment in the parks and other public places is of good musical quality and furnished free or at popular prices, is a great help in developing a critical attitude even among those in other respects quite uneducated. Therefore, we may say with great satisfaction that training in the art of hearing music has come upon the educational horizon in America in a very promising way.

The scientific attitude. We are constantly impressed with the analogy between scientific insight into the nature of music and the corresponding insight in other sciences, such as botany or astronomy. A child starts out with an inborn capacity and urge for the love of flowers; but as a student of flowers he soon encounters many marvelous revelations. On the one hand his horizon is broadened by acquaintance with larger and larger classes and varieties. He learns to see relationships to habitat. He learns to trace scientifically the laws of their evolution, the methods of artificial breeding and development of new plants, and even the beauty in their usefulness. On

the other hand, he turns to his microscope and discovers not only the external parts visible to the naked eye but the internal structures, their modes of evolution and development, even down to the discovery of the mechanisms of heredity through the genes, which are so small that we know them only by inference from what the microscope reveals. The natural history of music is analogous to that of such a material science.

This scientific insight into the structure and function of music goes hand in hand with the development of the artistic insight into the nature of the esthetic values from the point of view of the art of music. Thus the student of music is now furnished the facilities for increasing his power of appreciation of music and a critical awarding of praise or blame from two complementary points of view: a study of the art and a study of the science of music.

Terminology. The mere matter of terminology here plays a very important role, as is now being demonstrated so successfully in the recent developments in the field of psychology of music and acoustics. Take, for example, the concept of tone quality. Until a few years ago there was probably not a single adequate statement of the nature of tone quality in the entire literature on music; and yet this is the most important element in music. In the past the listener and the student have generally approached the subject in an attitude of acceptance or rejection of the unanalyzed impression, but often with no more competence than that with which the ignorant shepherd appraises the grandeur and meaning of the heavens above on a starlit night. The teacher, as a rule, said "This is beautiful," or "This is ugly," without being able to give the reasons why, for want of a tangible terminology. The teaching and development of tone quality therefore has wallowed in a slough of despond. Recent researches in the field of musical acoustics are ushering in a new era. The structure of tone quality has been dissected, and its parts have been adequately described and defined. This is leading to an improved musical language. The same is true for other musical concepts. Thus, through the systematic development of musical terminology, there will gradually develop a scientific classification of the various aspects of the musical medium which is essential not only for the learning of music but also for the awarding of praise and blame.

Musical talent. With this development of a knowledge of the nature of music, the scientific student of music has turned to the analysis of the nature of the musical mind attempting to assign specific roles to musical

heredity and musical training and environment. Hazy concepts of the gift of music have been subjected to scientific analysis, experiment, and measurement. From this, it is being demonstrated that it is impossible to award praise or blame without taking a reasonable account of the innate fortes and faults of the performer which determine in large part the direction and limits of achievement or failure. The recognition of this fact, of course, plays its fundamental role in the early encouragement or discouragement of the inceptive musician, in the recognition of the purpose of his education, and in the motivation through training.

PARTIES CONCERNED

The pupil. Modern education has revealed the importance of a student-centered and student-originated learning process as opposed to the traditional and institutional direction of development of the individual. The question of self-praise and self-blame, satisfaction or dissatisfaction on the part of the child or youth, is of course of first importance both in the initial stages of orientation and throughout the course of specialization and achievement. Instead of imposing regimentation, the home, the school and the private teachers tactfully aim to assist the child in the discovery of his natural interests, resources and promises which aid him in the building of his personal convictions. But emphasis is laid upon the principle of self-determination in the light of progressively available reasonable facilities for self-appraisal. This refers both to the appreciation of music and to the development of musical skills. This principle of self-determination has frequently run amuck in the hands of progressive educators, capital *P*. But it is a fundamental and permanent principle for the development of personality and motivation in specific pursuits. The child will soon discover that in order to pursue his interests and satisfy himself in performance he may have to work hard, conform to requirements, be patient, and make sacrifices; but his first goal is to feel satisfaction in what he is getting and in the worth-whileness of the pursuit.

For these things he will find an analogy in any form of play, physical or mental, into which he enters in a wholehearted way. This feeling of self-approbation comes easily in the musically talented where the urge for music is clearly manifested; but with moderate talent this attitude of self-reliance needs cultivation. And in the assumed absence of talent, there is need of assistance in the real verification of this absence. The greatest danger lies in the intrusion of prejudice and social pressures. This principle of self-determination is of most importance in those fields which are to be pursued primarily as a source of self-expression and pleasure in the pursuit as distinguished from the pursuits which are essential for the earning of bread and butter and the maintaining of social status.

The teacher. The recognition of this principle of self-determination and the sustained will to achieve has revolutionized the art of teaching, or we

may say better, has laid the beginnings for an art of teaching. One of the greatest obstacles the teacher encounters is the traditional social demand, and often the recognized objective in the profession, to make everybody musical. Finding that there is no adequate natural interest and no marks of gratifying achievement are developing, the teacher settles down to the sledge-hammer method of forced development. We must therefore bespeak for the live music teacher the right to refuse to make gold out of iron. The private teacher's fee and the tuition of the school are, of course, the temptations of the devil to face the situation with complacency. The constituency granting lenience in this respect has the right to expect that, in turn, the teacher understands and has exercised the effective means of self-discovery, self-orientation, self-determination, and self-motivation on the part of those who are on the border line. In this art, the profession has made but little progress so far. The constituency also has the right to impose the obligation upon the teacher to respect the individuality of the gifted and to give all types of freedom to the pupil commensurate with his natural line and degree of successful achievement in developing appreciation for various types of music.

In breaking with the traditional routines of musical drill, the teacher must first of all give up the attempt to cast the musical mind of the child in the mold of his or her own pattern. This principle has acquired its greatest significance in dealing with the highly gifted, where it is clearly substantiated by the musical history of precocious children. The public must acquire tolerance for the principle of giving musical facilities to child and youth in proportion to this self-determination and demonstrated ability to achieve. The great musician and the musical virtuoso are not fostered by educational or social conformance but primarily by the enjoyment of the principle of free determination and development of divergent personality.

The critic. The music critic unfortunately, as a rule, is a newspaper man and must conform to the pressure methods of efficiency in writing down. This newspaper ability has often been the basis for the unfortunate selection of musical critics. On the other hand, the thoroughly competent musical critic, if he were to ply his art as a technician, would find but small outlet for publication and a negligible constituency of readers. He would face, most discouragingly, the fact that many of the performers on the stage do not have critical knowledge of what they are doing and are not interested or

even capable of reading the highly technical and critical analysis of their performance. So we must be tolerant with the musical critic.

If music were a simple thing or had a specific goal, such as the development of a logical proposition, the task of the critic would be easier. But, as it is, he is dealing with a highly fluid and chaotic state of affairs when he is supposed to aid the common reader in assigning praise or blame to the performance. One of the most striking faults of the work of the musical critic is his lack of words—lack of concepts which are discriminating and logically definable. We tolerate his splashing of his personal likes or dislikes as if he were competent to boil down our feelings of the situation into a fair assaying of praise or blame of the performance. One difficulty lies in the demand for writing in smart and emotional style. Among critics there are rare individuals who have a fine artistic insight and a balanced esthetic judgment in regard to what is good or bad in music. But we seldom see their names in the musical-review columns.

Phonophotographic recordings reveal art principles which had not been discovered in the musical world through hearing, and yet functioned very largely. They enable the critic to use scientific language in the description of achievement and assign praise or blame on the basis of identifiable and describable grounds. We can now photograph the sound waves of a performance at any distance from the source of sound and convert this into an analyzable performance score. As a whole, this principle cannot be available to the musical critic who must render an immediate opinion. But we have a right to expect that he should have a scientific knowledge of the types of facts which are revealed in such objective records. He is living in a new musical era in which the qualifications for the critic are as new as they are for the artistic and scientific orientation of teacher and performer.

The public. I can recall the time when the possession of a piano and piano lessons for the darling girl represented almost the sole ambition of the mothers in our state aside from the spontaneous and untutored self-expression in song. The public demand has changed. Instruments have changed. Methods of teaching have changed. Understanding of the art has increased. Cosmopolitan tastes have spread. Knowledge of the science of music has become aligned with knowledge in other sciences. The public rapidly learns to demand all these things and take the right for granted. The ambition of the mother, the objective of the teacher, and the expression of the youth of fifty years ago are coming to be historical curiosities. The art

of music in America is coming to a wholesome fruition and is becoming associated with present-day academic learning. For the first time in history it reaches out into the remotest corners of the land and many of the technical aspects of the art and new concepts of the musical medium are fast becoming common-sense knowledge and therefore add in the just awarding of praise or blame.

THOUGHT REVIEW

Principles

- (1) Music is designed to give pleasure when heard, and therefore calls for immediate affective response.
- (2) Capacity for esthetic response varies with natural talent and training, both artistic and scientific.
- (3) It is important to know "where to laugh and when to laugh."
- (4) For scientific description of likes or dislikes we need a discriminating terminology.
- (5) As every scientific discovery and invention gradually becomes common-sense knowledge when popularly known, so artistic judgment tends to crystallize into common-sense feeling.
- (6) Musical criticism may be chastized and chastened into balanced and critical judgment, relatively free from emotional explosions.
- (7) The spontaneous and natural smile or frown is often more eloquently telling or cutting than words.

Consider These Questions

- (1) Is the conventional response to music affectedly temperamental?
- (2) Why is listening intently to a major musical performance more exhausting than listening to an equally engrossing lecture on "International Relations?"
- (3) Was the English visitor right when he said, "You have no art, because whatever you have you display"?
- (4) Does insight into the nature and structure of the thing beautiful tend to lessen or to strengthen the artistic appeal?

Discuss This Situation

- (1) During the last twenty years in a Midwestern city of 30,000 people music has "come to town." Curricular and extracurricular music in the schools has been tenfold what it was during the foregoing twenty years. Music clubs and other musical activities have greatly increased. The community often takes as strong an interest in local, district, state, and national contests in music as ordinarily attaches to football. Discuss the effect of this upon (a) the musical, (b) the social, (c) the recreational and (d) the educational life of the community. Evaluate the change in interest and capacity for assigning fair praise and blame to music.

Transcriber's Note

Where hyphenation occurs on a line break, the decision to retain or remove is based on occurrences elsewhere in the text.

The errors deemed most likely to be the printer's have been corrected, and are noted here. The references are to the page and line in the original text.

8.2	Play is essentially social and find[s] its highest	added
21.13	The old conflict be[t]ween enthusiasts for rote singing	added
38.11	has, of course, been unparall[el]ed in the history of the world.	added
39.48	generally has a[n] hereditary basis in	removed
66.30	for example, in the present "Novac[h]ord"	added
71.23	flexibil[i]ty of tone quality is even more desirable.	added

*** END OF THE PROJECT GUTENBERG EBOOK WHY WE LOVE
MUSIC ***

Updated editions will replace the previous one—the old editions will be renamed.

Creating the works from print editions not protected by U.S. copyright law means that no one owns a United States copyright in these works, so the Foundation (and you!) can copy and distribute it in the United States without permission and without paying copyright royalties. Special rules, set forth in the General Terms of Use part of this license, apply to copying and distributing Project Gutenberg™ electronic works to protect the PROJECT GUTENBERG™ concept and trademark. Project Gutenberg is a registered trademark, and may not be used if you charge for an eBook, except by following the terms of the trademark license, including paying royalties for use of the Project Gutenberg trademark. If you do not charge anything for copies of this eBook, complying with the trademark license is very easy. You may use this eBook for nearly any purpose such as creation of derivative works, reports, performances and research. Project Gutenberg eBooks may be modified and printed and given away—you may do practically ANYTHING in the United States with eBooks not protected by U.S. copyright law. Redistribution is subject to the trademark license, especially commercial redistribution.

START: FULL LICENSE

THE FULL PROJECT GUTENBERG™ LICENSE

PLEASE READ THIS BEFORE YOU DISTRIBUTE OR USE THIS WORK

To protect the Project Gutenberg™ mission of promoting the free distribution of electronic works, by using or distributing this work (or any other work associated in any way with the phrase “Project Gutenberg”), you agree to comply with all the terms of the Full Project Gutenberg License available with this file or online at www.gutenberg.org/license.

Section 1. General Terms of Use and Redistributing Project Gutenberg electronic works

1.A. By reading or using any part of this Project Gutenberg electronic work, you indicate that you have read, understand, agree to and accept all the terms of this license and intellectual property (trademark/copyright) agreement. If you do not agree to abide by all the terms of this agreement, you must cease using and return or destroy all copies of Project Gutenberg electronic works in your possession. If you paid a fee for obtaining a copy of or access to a Project Gutenberg electronic work and you do not agree to be bound by the terms of this agreement, you may obtain a refund from the person or entity to whom you paid the fee as set forth in paragraph 1.E.8.

1.B. “Project Gutenberg” is a registered trademark. It may only be used on or associated in any way with an electronic work by people who agree to be bound by the terms of this agreement. There are a few things that you can do with most Project Gutenberg electronic works even without complying with the full terms of this agreement. See paragraph 1.C below. There are a lot of things you can do with Project Gutenberg electronic works if you follow the terms of this agreement and help preserve free future access to Project Gutenberg electronic works. See paragraph 1.E below.

1.C. The Project Gutenberg Literary Archive Foundation (“the Foundation” or PGLAF), owns a compilation copyright in the

collection of Project Gutenberg electronic works. Nearly all the individual works in the collection are in the public domain in the United States. If an individual work is unprotected by copyright law in the United States and you are located in the United States, we do not claim a right to prevent you from copying, distributing, performing, displaying or creating derivative works based on the work as long as all references to Project Gutenberg are removed. Of course, we hope that you will support the Project Gutenberg mission of promoting free access to electronic works by freely sharing Project Gutenberg works in compliance with the terms of this agreement for keeping the Project Gutenberg name associated with the work. You can easily comply with the terms of this agreement by keeping this work in the same format with its attached full Project Gutenberg License when you share it without charge with others.

1.D. The copyright laws of the place where you are located also govern what you can do with this work. Copyright laws in most countries are in a constant state of change. If you are outside the United States, check the laws of your country in addition to the terms of this agreement before downloading, copying, displaying, performing, distributing or creating derivative works based on this work or any other Project Gutenberg work. The Foundation makes no representations concerning the copyright status of any work in any country other than the United States.

1.E. Unless you have removed all references to Project Gutenberg:

1.E.1. The following sentence, with active links to, or other immediate access to, the full Project Gutenberg License must appear prominently whenever any copy of a Project Gutenberg work (any work on which the phrase “Project Gutenberg” appears, or with which the phrase “Project Gutenberg” is associated) is accessed, displayed, performed, viewed, copied or distributed:

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg™ License included with

this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

1.E.2. If an individual Project Gutenberg electronic work is derived from texts not protected by U.S. copyright law (does not contain a notice indicating that it is posted with permission of the copyright holder), the work can be copied and distributed to anyone in the United States without paying any fees or charges. If you are redistributing or providing access to a work with the phrase “Project Gutenberg” associated with or appearing on the work, you must comply either with the requirements of paragraphs 1.E.1 through 1.E.7 or obtain permission for the use of the work and the Project Gutenberg trademark as set forth in paragraphs 1.E.8 or 1.E.9.

1.E.3. If an individual Project Gutenberg electronic work is posted with the permission of the copyright holder, your use and distribution must comply with both paragraphs 1.E.1 through 1.E.7 and any additional terms imposed by the copyright holder. Additional terms will be linked to the Project Gutenberg License for all works posted with the permission of the copyright holder found at the beginning of this work.

1.E.4. Do not unlink or detach or remove the full Project Gutenberg License terms from this work, or any files containing a part of this work or any other work associated with Project Gutenberg.

1.E.5. Do not copy, display, perform, distribute or redistribute this electronic work, or any part of this electronic work, without prominently displaying the sentence set forth in paragraph 1.E.1 with active links or immediate access to the full terms of the Project Gutenberg License.

1.E.6. You may convert to and distribute this work in any binary, compressed, marked up, nonproprietary or proprietary form, including any word processing or hypertext form. However, if you provide access to or distribute copies of a Project Gutenberg work in a format other than “Plain Vanilla ASCII” or other format used in the official

version posted on the official Project Gutenberg website (www.gutenberg.org), you must, at no additional cost, fee or expense to the user, provide a copy, a means of exporting a copy, or a means of obtaining a copy upon request, of the work in its original “Plain Vanilla ASCII” or other form. Any alternate format must include the full Project Gutenberg License as specified in paragraph 1.E.1.

1.E.7. Do not charge a fee for access to, viewing, displaying, performing, copying or distributing any Project Gutenberg works unless you comply with paragraph 1.E.8 or 1.E.9.

1.E.8. You may charge a reasonable fee for copies of or providing access to or distributing Project Gutenberg electronic works provided that:

- You pay a royalty fee of 20% of the gross profits you derive from the use of Project Gutenberg works calculated using the method you already use to calculate your applicable taxes. The fee is owed to the owner of the Project Gutenberg trademark, but he has agreed to donate royalties under this paragraph to the Project Gutenberg Literary Archive Foundation. Royalty payments must be paid within 60 days following each date on which you prepare (or are legally required to prepare) your periodic tax returns. Royalty payments should be clearly marked as such and sent to the Project Gutenberg Literary Archive Foundation at the address specified in Section 4, “Information about donations to the Project Gutenberg Literary Archive Foundation.”
- You provide a full refund of any money paid by a user who notifies you in writing (or by e-mail) within 30 days of receipt that s/he does not agree to the terms of the full Project Gutenberg™ License. You must require such a user to return or destroy all copies of the works possessed in a physical medium and discontinue all use of and all access to other copies of Project Gutenberg™ works.
- You provide, in accordance with paragraph 1.F.3, a full refund of any money paid for a work or a replacement copy, if a defect in the electronic work is discovered and reported to you within 90 days of receipt of the work.

- You comply with all other terms of this agreement for free distribution of Project Gutenberg™ works.

1.E.9. If you wish to charge a fee or distribute a Project Gutenberg™ electronic work or group of works on different terms than are set forth in this agreement, you must obtain permission in writing from the Project Gutenberg Literary Archive Foundation, the manager of the Project Gutenberg™ trademark. Contact the Foundation as set forth in Section 3 below.

1.F.

1.F.1. Project Gutenberg volunteers and employees expend considerable effort to identify, do copyright research on, transcribe and proofread works not protected by U.S. copyright law in creating the Project Gutenberg™ collection. Despite these efforts, Project Gutenberg™ electronic works, and the medium on which they may be stored, may contain “Defects,” such as, but not limited to, incomplete, inaccurate or corrupt data, transcription errors, a copyright or other intellectual property infringement, a defective or damaged disk or other medium, a computer virus, or computer codes that damage or cannot be read by your equipment.

1.F.2. LIMITED WARRANTY, DISCLAIMER OF DAMAGES - Except for the “Right of Replacement or Refund” described in paragraph 1.F.3, the Project Gutenberg Literary Archive Foundation, the owner of the Project Gutenberg™ trademark, and any other party distributing a Project Gutenberg™ electronic work under this agreement, disclaim all liability to you for damages, costs and expenses, including legal fees. YOU AGREE THAT YOU HAVE NO REMEDIES FOR NEGLIGENCE, STRICT LIABILITY, BREACH OF WARRANTY OR BREACH OF CONTRACT EXCEPT THOSE PROVIDED IN PARAGRAPH 1.F.3. YOU AGREE THAT THE FOUNDATION, THE TRADEMARK OWNER, AND ANY DISTRIBUTOR UNDER THIS AGREEMENT WILL NOT BE LIABLE TO YOU FOR ACTUAL, DIRECT, INDIRECT, CONSEQUENTIAL, PUNITIVE OR INCIDENTAL DAMAGES

EVEN IF YOU GIVE NOTICE OF THE POSSIBILITY OF SUCH DAMAGE.

1.F.3. LIMITED RIGHT OF REPLACEMENT OR REFUND - If you discover a defect in this electronic work within 90 days of receiving it, you can receive a refund of the money (if any) you paid for it by sending a written explanation to the person you received the work from. If you received the work on a physical medium, you must return the medium with your written explanation. The person or entity that provided you with the defective work may elect to provide a replacement copy in lieu of a refund. If you received the work electronically, the person or entity providing it to you may choose to give you a second opportunity to receive the work electronically in lieu of a refund. If the second copy is also defective, you may demand a refund in writing without further opportunities to fix the problem.

1.F.4. Except for the limited right of replacement or refund set forth in paragraph 1.F.3, this work is provided to you 'AS-IS', WITH NO OTHER WARRANTIES OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO WARRANTIES OF MERCHANTABILITY OR FITNESS FOR ANY PURPOSE.

1.F.5. Some states do not allow disclaimers of certain implied warranties or the exclusion or limitation of certain types of damages. If any disclaimer or limitation set forth in this agreement violates the law of the state applicable to this agreement, the agreement shall be interpreted to make the maximum disclaimer or limitation permitted by the applicable state law. The invalidity or unenforceability of any provision of this agreement shall not void the remaining provisions.

1.F.6. INDEMNITY - You agree to indemnify and hold the Foundation, the trademark owner, any agent or employee of the Foundation, anyone providing copies of Project Gutenberg™ electronic works in accordance with this agreement, and any volunteers associated with the production, promotion and distribution of Project Gutenberg™ electronic works, harmless from all liability, costs and expenses, including legal fees, that arise directly or indirectly from any of the following which you do or cause to occur: (a)

distribution of this or any Project Gutenberg work, (b) alteration, modification, or additions or deletions to any Project Gutenberg work, and (c) any Defect you cause.

Section 2. Information about the Mission of Project Gutenberg

Project Gutenberg is synonymous with the free distribution of electronic works in formats readable by the widest variety of computers including obsolete, old, middle-aged and new computers. It exists because of the efforts of hundreds of volunteers and donations from people in all walks of life.

Volunteers and financial support to provide volunteers with the assistance they need are critical to reaching Project Gutenberg's goals and ensuring that the Project Gutenberg collection will remain freely available for generations to come. In 2001, the Project Gutenberg Literary Archive Foundation was created to provide a secure and permanent future for Project Gutenberg and future generations. To learn more about the Project Gutenberg Literary Archive Foundation and how your efforts and donations can help, see Sections 3 and 4 and the Foundation information page at www.gutenberg.org.

Section 3. Information about the Project Gutenberg Literary Archive Foundation

The Project Gutenberg Literary Archive Foundation is a non-profit 501(c)(3) educational corporation organized under the laws of the state of Mississippi and granted tax exempt status by the Internal Revenue Service. The Foundation's EIN or federal tax identification number is 64-6221541. Contributions to the Project Gutenberg Literary Archive Foundation are tax deductible to the full extent permitted by U.S. federal laws and your state's laws.

The Foundation's business office is located at 41 Watchung Plaza #516, Montclair NJ 07042, USA, +1 (862) 621-9288. Email contact

links and up to date contact information can be found at the Foundation's website and official page at www.gutenberg.org/contact

Section 4. Information about Donations to the Project Gutenberg Literary Archive Foundation

Project Gutenberg™ depends upon and cannot survive without widespread public support and donations to carry out its mission of increasing the number of public domain and licensed works that can be freely distributed in machine-readable form accessible by the widest array of equipment including outdated equipment. Many small donations (\$1 to \$5,000) are particularly important to maintaining tax exempt status with the IRS.

The Foundation is committed to complying with the laws regulating charities and charitable donations in all 50 states of the United States. Compliance requirements are not uniform and it takes a considerable effort, much paperwork and many fees to meet and keep up with these requirements. We do not solicit donations in locations where we have not received written confirmation of compliance. To SEND DONATIONS or determine the status of compliance for any particular state visit www.gutenberg.org/donate.

While we cannot and do not solicit contributions from states where we have not met the solicitation requirements, we know of no prohibition against accepting unsolicited donations from donors in such states who approach us with offers to donate.

International donations are gratefully accepted, but we cannot make any statements concerning tax treatment of donations received from outside the United States. U.S. laws alone swamp our small staff.

Please check the Project Gutenberg web pages for current donation methods and addresses. Donations are accepted in a number of other ways including checks, online payments and credit card donations. To donate, please visit: www.gutenberg.org/donate.

Section 5. General Information About Project Gutenberg electronic works

Professor Michael S. Hart was the originator of the Project Gutenberg concept of a library of electronic works that could be freely shared with anyone. For forty years, he produced and distributed Project Gutenberg eBooks with only a loose network of volunteer support.

Project Gutenberg eBooks are often created from several printed editions, all of which are confirmed as not protected by copyright in the U.S. unless a copyright notice is included. Thus, we do not necessarily keep eBooks in compliance with any particular paper edition.

Most people start at our website which has the main PG search facility: www.gutenberg.org.

This website includes information about Project Gutenberg, including how to make donations to the Project Gutenberg Literary Archive Foundation, how to help produce our new eBooks, and how to subscribe to our email newsletter to hear about new eBooks.